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Operations and regulations for a ride-sourcing market with a mixed fleet of human drivers and autonomous vehicles

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ABSTRACT

The rise of autonomous driving may sharply reshape ride-sourcing markets by influencing drivers' income and participation, customers' choices, ride-sourcing platforms' operations, and governments' regulations. We introduce a model that characterizes a ride-sourcing market with a mixed fleet of non-atomic autonomous vehicles (AVs) and atomic human-driven vehicles (HVs). Human drivers decide whether and where to offer ride-sourcing services independently, while AVs are fully controlled and dispatched by the platform to serve on-demand ride requests that are distributed heterogeneously. We solve for the optimal platform operating strategy and examine the implications of different government regulations on the platform's optimal decisions and the resulting market equilibrium. Although the emergence of AV might undermine human drivers' welfare, it can simultaneously increase platform profit, customer surplus, and social welfare. Moreover, by looking into the regulatory outcomes of a few prevailing regulations (i.e., price cap, minimum wage, minimum fleet size, and maximum fleet size), we find that the policymakers should determine the regulations according to the conditions of labor supply and the development stage of AV technology.

1. Introduction

The rapid development of intelligent transportation has catalyzed the rise of autonomous vehicles (AVs) (Mahmassani, 2016). The concept of AVs has been changing over time. According to a five-level classification introduced by the Society of Automotive Engineers International,[1] the development of a driving automation system is from driver assistance in level 1 to full driving automation in level 5; fully mature autonomous vehicles can ride on public roads without human drivers' assistance. Developed AVs are expected to be operated as both private and commercial vehicles and have the potential to improve public transportation and company/platform benefits. For urban administrators or planners, introducing an AV fleet can better relieve traffic congestion and improve safety (Pomerleau and Jochem, 1996), because AVs are equipped with advanced sensing and control systems, and their road behavior shows strong controllability and low uncertainty.[2] Private car owners can also benefit greatly from AVs. Well-designed automotive cruise system (Lu et al., 2019) in AVs effectively reduces energy consumption (Faisal et al., 2019), and drivers can experience comfortable drives. From the company/platform's standpoint, the AV fleet will promote collaborative mobility, such as shared autonomous vehicles (SAVs), which can provide less expensive ride services for passengers. In addition, good information connectivity and controllability (Pourmehrab et al., 2019) of AVs can help the platform maximize its profit by following the

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[1] Source: https://www.sae.org/standards (accessed in November 2023).

[2] It should be noted that the advance of AV technology might worsen traffic congestion due to rebounding effects (Small and Verhoef, 2007).

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group's instructions (Shladover, 2018). KPMG[3] measures the level of preparedness for autonomous vehicles across 30 countries and jurisdictions. Some cities have already hosted AV pilot operations, and others are preparing long-term plans for AVs. Both city government and AV operators play roles in these pilots. Besides, many commercial companies, like Volvo, General Motors, TESLA, etc., have paid much attention to the design of their autonomous products. It is predicted that the AV market share will reach as high as 87.2% by the year 2045 (Bansal and Kockelman, 2017).

In particular, autonomous driving may significantly reshape the on-demand ride-sourcing service, which is a symbolic icon of shared mobility and a popular mobility mode among travelers. The ride-sourcing service efficiently connects customers who request on-demand riders with drivers who provide ride services through digital platforms, such as Uber, Lyft, and Didi. Such platforms are generally termed *transportation network companies* (TNCs). Although most TNCs rely on human drivers to operate their services in the current stage, recent years have witnessed a trend of using AV fleets to satisfy customers' on-demand mobility needs. For example, Waymo was established under Alphabet and began to provide its self-driving service in Phoenix in 2018, which is the world's first commercial autonomous ride-hailing service. Besides, well-known ride-sourcing companies like Uber[4] and DiDi[5] also started to establish their autonomous vehicle fleet for providing on-demand ride-hailing services.

Heated discussions are made by researchers, media, and the public on the far-reaching implications of the widespread use of AVs in ride-sourcing markets. Milakis et al. (2017) suggest that the adoption of SAVs could potentially replace a substantial number (from around 67% to 90%) of conventional vehicles, partially due to the relatively cheap charge in SAV service. On the other hand, human drivers working for ride-sourcing services may suffer from the loss of customers, the reduction of expected income, and even the potential of losing their jobs, as AVs penetrate ride-sourcing markets. From the viewpoint of urban administrators and planners, comfort, reliability, and reduced perceived value of time brought by AVs may lead to long commute distances, contributing to urban sprawl. Additionally, some investigations show that AVs can reduce traffic congestion only by using platooning technology. The building of dedicated lanes/regions for AV operation also results in enormous construction costs (Faisal et al., 2019). In summary, the introduction of AVs into ride-sourcing markets may raise great challenges for governments to adjust their regulatory policies for ride-sourcing services.

Some studies indicate that a deregulated ride-sourcing market can result in monopolistic competitions (Ch et al., 2006) or even non-existence of equilibrium (Cairns and Liston-Heyes, 1996). In order to obtain the first-best result, regulations should be carefully considered case by case, which include subsidization (Arnott, 1996), price, entry (Salanova et al., 2011), etc. In reality, some strict regulatory policies have been imposed for ride-sourcing services provided by human drivers in cities, particularly metropolitan areas. For example, in major cities in China, such as Beijing and Shanghai, only residents with local licenses are allowed to provide ride-sourcing services. Similarly, only drivers with a for-hire-vehicle (FHV) license are permitted to offer on-demand rides to customers in New York City (NYC). These rules are essentially means of fleet size control, which aims at avoiding the overcrowding of ride-sourcing vehicle fleets from increasing traffic congestion and exhaust gas emissions. NYC has additionally issued other strict regulations, including the minimum wage guarantee for drivers, and minimum utilization rate control (TNCs must ensure that the utilization rate of their affiliated vehicles is above a certain level), to protect labor welfare and alleviate traffic clogs. However, these regulations designed for markets with only human drivers may not necessarily achieve the desirable socially beneficial outcomes in the era of autonomous driving. For example, the minimum wage guarantee, which is intended to protect drivers' welfare, may be counter-productive since the ride-sourcing platforms may decide to invest in AV fleets instead of recruiting human drivers. It is thus of great interest to investigate the implications of AV adoption on the equilibrium state of ride-sourcing markets, examine the platform operating strategy (in terms of price, wage, and AV fleet provision), and design appropriate government regulatory schemes for enhancing social welfare in ride-sourcing markets with a mixed fleet of human drivers and AVs. Our main research questions are as follows: (1) Can autonomous driving contribute to both platform and society? (2) Will autonomous driving influence the utility of human drivers? (3) How should the government/policymaker regulate the mixed market?

To address these issues, this paper makes one of the initial attempts to model a ride-sourcing market under a mixed autonomous paradigm. Human drivers are atomic and try to optimize their profits by deciding whether to participate in the platform and which region to offer ride services. A Wardrop equilibrium is achieved when no human driver can improve his/her utility by unilaterally altering his/her decision. In contrast, the AV fleet is fully controlled by the ride-sourcing platform, which directly determines the AV fleet size and distribution of AV fleets in different regions. Based on such a model, we solve for the analytical solutions of monopoly optimum and social optimum, and discuss their properties. Furthermore, we examine the platform's optimal operating strategy under different regulations imposed by the government (i.e., price-cap, minimum wage, minimum fleet size, and maximum fleet size), from which we can understand the implications of these regulations on the platform's decisions and the resulting market equilibrium.

Our model identifies the distinct marginal cost between AVs and human-driven vehicles (HVs) as the key driver in influencing the platform's decisions. The marginal cost of recruiting human drivers increases with the number, while the cost of deploying AVs remains constant. This intrinsic difference results in inevitable competition among different servers and justifies the platform's adoption strategy toward emerging technologies. Specifically, our study offers a few interesting managerial insights and suggestions for ride-sourcing companies and policy-makers to better operate and regulate ride-sourcing services with the AV fleet, respectively. We have the following main findings. (1) Once the technology of AV is mature enough, its penetration in ride-sourcing markets

[3] Source: https://kpmg.com/xx/en/home/insights/2020/06 (accessed in November 2023).
[4] Source: https://www.cnbc.com/2020/01/28 (accessed in November 2023).
[5] Source: https://www.reuters.com/article (accessed in November 2023).

can improve platform welfare and customer surplus but inevitably worsen labor welfare. (2) The advance of AV technology can benefit the platform and society; however, it widens the gap of social welfare between the monopoly and social optimums. The improvement of AV is more beneficial to the platform than society because the platform is the one to deploy the AV fleet. (3) The government can establish regulations to induce the dominant platform to improve social welfare, such as restrictions on AV's fleet size or minimum wage. In contrast, improper regulation, like restricting the maximum number of service providers or putting too many AVs with expensive maintenance costs into the market, harms both the government and platforms.

This paper is organized as follows. Section 2 reviews some previous studies on autonomous driving and regulations on ride-sourcing markets. Section 3 illustrates the basic model describing a hybrid market. Section 4 describes the equilibrium outcomes under different market structures. Section 5 examines how the penetration of AVs influences human drivers and society, and further explores how the government can conduct regulations to improve social welfare. Section 6 studies an example of AV fleet application in relocation problems. Section 7 summarizes the conclusions and proposes some future work.

2. Literature review

Our study focuses on the impact of AVs on the traditional ride-sourcing market and how appropriate regulations can be designed for the future ride-sourcing market with a mixed autonomous paradigm. Therefore, we mainly reviewed the literature from two aspects: (1) operations and regulations for ride-sourcing markets and (2) implications of AVs in transportation systems and regulations for AVs.

In the traditional ride-sourcing market, regulations like price caps, wage standards, and others have been investigated. Zha et al. (2016) show that, by capping the commission charged by the ride-sourcing platforms, the government can achieve a second-best solution that can maximize social welfare and ensure non-negative platform profit. Li et al. (2019) evaluate three proposed regulations, i.e., the minimum wage, the cap on the vehicle number, and the congestion tax, and discuss the potential impact on the benefit of both drivers and passengers. Hu and Zhou (2020) consider the regulatory methods of price and wage for an on-demand service platform, and they mainly investigate the performance of the fixed commission under various market conditions. Yu et al. (2020) examine some policies implemented by the Chinese government and the New York City Council. The findings show that, rather than no policy and complete ban regulations, a carefully designed regulatory policy can better benefit multiple competing objectives and total social welfare. Jiang et al. (2020) combine behavioral modeling and controlled experiments. They show that, when designing policies, on-demand platforms should attach importance to workers' behavioral biases, such as regret aversion. Vignon et al. (2021) investigate the effects of the regulatory schemes imposed by the NYC government by establishing an equilibrium model that considers product differentiation and traffic congestion externality. Ke et al. (2021) investigate the regulations of congested ride-sourcing markets under the context of heterogeneous demand and supply; they indicate that the regulations of commission regulation and demand regulation can be Pareto-efficient. Liu et al. (2022) investigate the regulations in electrified ride-sourcing markets; the platform's best response to three potential regulatory policies (annual permit fees, differential trip-based fees, and differential commission caps) are examined. Zhang and Nie (2022) examine how regulations can relieve traffic congestion in an idealized city that contains a dense central core surrounded by a larger periphery. Specifically, the implications of three policies (trip-based,cordon-based, and cruising cap) are discussed. Vignon et al. (2023) discuss how to regulate the competition between traditional street-hail and emerging ride-hailing services; they observe that the regulation of commission cap can replicate the second-best in this market.

In terms of the implications of AVs in transportation systems, an in-depth analysis should be conducted before making regulations on AVs (Asmussen et al., 2020). The transportation infrastructure (Zhang et al., 2020), energy consumption, and environmental impact (Jansuwan et al., 2021) can significantly influence the smooth operation and deployment of AVs. Besides, the likely benefit of AVs is associated with the proportions of AV and HV (Qin and Wang, 2019). From the perspective of both consumers and human drivers, the acceptance of vehicle-sharing (Liu et al., 2020) and drivers' behaviors towards AVs should also be considered (Zhao et al., 2020). In general, a complete analysis of AV's potential impact is necessary for policy-makers and operating platforms. As for specific regulation methods, Zhang et al. (2019) and Millard-Ball (2019) study the situation people daily drive with AVs. They highlight the differences in commuting and parking problems with or without AVs, and discuss the management of parking and traffic in different cases. Amirgholy et al. (2020) consider travel time and safety problems, and reveal that it is necessary to design and regulate the reserved lane for AVs. Luo et al. (2019) investigate the subsidy policy that accelerates the deployment of AVs and show the policy can maximize the expected pay-off of the government. Tscharaktschiew and Evangelinos (2019) consider the relationship between the chosen level of autonomous driving and the travel cost, and study how to adjust road congestion pricing when AV becomes a choice for drivers. Some up-to-date studies focus on city and society levels—they discuss the possible adverse influence brought by AVs and put forward corresponding suggestions (Kassens-Noor et al., 2020). Wang et al. (2021) investigate a crowdsourcing service of AVs where AV owners can rent their cars to a company. This innovative service shows great potential in benefiting social welfare or crowdsourcing platforms' profit.

Similar to our work, some recent studies discuss the penetration of AVs into ride-sourcing markets. Ahadi et al. (2023) consider the charging problem of shared AVs. In the presence of uncertain demands, time-varying electricity prices, and other conditions, they propose learning smart charging policies regarding when and where to charge vehicles. Gao and Li (2023) investigate the competition between the ride-hailing platform operating AV fleet and the public transit agency offering transit services. Interestingly, in the transportation network, they show that the introduction of AVs will enlarge the existing spatial and social inequity gaps. In general, our study differs from the above ones in the sense that we incorporate AVs into hybrid ride-sourcing markets and emphasize the interactions between AVs and traditional HVs. In this light, our work is most closely related to the following studies. Ao et al.

(2021) focus on a spatial problem in hybrid markets; they show that the platform prioritizes AV deployment in high-demand areas to make a higher profit, which pushes human drivers to suburbs, worsens earning opportunities for human drivers, and enlarges spatial inequity for passengers. Siddiq and Taylor (2022) discuss two operation patterns (platform-owned and individually-owned AVs) in hybrid markets. They indicate that the AV cost, the relative price sensitivity of demand, and the relative wage sensitivity of labor are critical in determining whether the platform's access to AVs can benefit different stakeholders. Lian and van Ryzin (2022) examine the equilibrium in a hybrid market under two dispatch patterns (common platform v.s. independent platform) and two levels of supply competition (monopoly AV v.s. competitive AV). Their main findings show that the advance of AVs can paradoxically increase the price charged for customers under some market structures. Mo et al. (2022) investigate a hybrid ride-sourcing market with the riders' perceived utility being heterogenous towards AVs and HVs; they indicate that the demand rates might not be monotonous to the price or fleet size. Generally, our work differs from these studies in examining the implications of different regulations in hybrid markets. Our study follows two processes. Firstly, we examine the equilibrium outcomes under different market structures and investigate how the AV penetration reshapes the ride-sourcing market, which largely echoes the findings in the existing literature. Furthermore, we discuss the implications of different regulations in hybrid markets and how policymakers should adopt distinct regulations based on specific conditions, which provides some managerial insights for the forthcoming hybrid rides-sourcing markets.

3. Basic model

This section proposes a model to delineate the mixed autonomous paradigm in the ride-sourcing market, which is formed by AVs and HVs. Consider a market with n regions, in which customers determine whether to use ride-sourcing services based on travel costs. Human drivers decide whether and where to serve customers depending on the expected income in different regions. In comparison, a fleet of autonomous drivers is provided and fully controlled by the ride-sourcing platform. The behaviors and characteristics of these stakeholders are elaborated on below.

Customers. The platform charges p for customers in all regions, which varies between 0 and 1. As for region $i \in N$, $\bar{D}_i$ and D_i denote the potential and effective demands, respectively. The effective demand D_i is negatively correlated with the price p:

$$D_i = \bar{D}_i(1 - p - hW_i), \tag{1}$$

where W_i is the expected waiting time in region i and h is the customer's delay sensitivity. S_i^h and S_i^a represent the numbers of HVs and AVs in region i. Following some previous literature (Mo et al., 2022; Benjaafar et al., 2022; Li et al., 2022), we assume the demand linearly decreases with price and waiting time, which enables us to derive tractable results. Our main insights are robust after incorporating more sophisticated forms of demand function. The primary goal of our study is to investigate how the introduction of AVs affects the market equilibrium in ride-sourcing systems. The main insights highlight the effects of the supply side on the platform and society, and do not depend on the specific demand form. Similar to previous work (Taylor, 2018; Bai et al., 2019; Benjaafar et al., 2022), we adopt the M/M/k queue to describe the waiting process. The waiting time is approximated as follows:

$$W_i = \frac{\rho_i^{\sqrt{2(k+1)}}}{\lambda_i(1 - \rho_i)}, \tag{2}$$

where λ_i is the rate of arriving customers in region i, measured by D_i in our model. ρ_i represents the utilization rate of the servers in region i, which is the ratio between the effective demand and the number of vehicles (S_i^h for HVs and S_i^a for AVs):

$$\rho_i = \frac{D_i}{S_i^h + S_i^a}. \tag{3}$$

To well define the delay process, we focus on the nontrivial equilibrium in demand and labor supply such that $S_i^h + S_i^a \geq D_i$. According to the statement in Bai et al. (2019), making k related to the number of suppliers leads to only iterative approaches, so k needs to be pre-defined to make it computationally feasible. Moreover, This study focuses on how a centralized platform forms its operation schemes, so we suppose the queue has only one server $k = 1$ to provide service. Under such circumstances, Eq. (2) has the exact form:

$$W_i = \frac{D_i/(S_i^h + S_i^a)}{S_i^h + S_i^a - D_i}. \tag{4}$$

By combining Eqs. (1) and (4), we can refine the effective demand in region i as follows:

$$D_i = \frac{1}{2}\Big[S_i^h + S_i^a - \sqrt{\left(S_i^h + S_i^a + \bar{D}_i(1-p) + \frac{h\bar{D}_i}{S_i^h + S_i^a}\right)^2 - 4\bar{D}_i(1-p)(S_i^h + S_i^a)} + \bar{D}_i(1-p) + \frac{h\bar{D}_i}{S_i^h + S_i^a}\Big]. \tag{5}$$

Autonomous vehicles. The platform fully controls the size and distribution of the AV fleet. S^a represents the total number of AVs, and S_i^a is the number of AVs in region i. The platform can adjust the ratio of AVs in these regions by γ_i, which is the proportion of AVs assigned to region i, and it is formulated as:

$$\gamma_i = S_i^a / S^a. \tag{6}$$

The platform is responsible for AV maintenance. All AVs put in the market need the fixed cost c_f, and only in-service AVs require the variable cost c_v.

Human vehicles. Unlike AVs, human drivers make decisions to maximize their utilities, and the HVs' distribution is determined based on the actual earnings. The platform sets wage w for human drivers for each trip. We assume there is a total number of $\bar{S}^h$ potential HV drivers in the market. Let S_i^h denote the effective numbers of HV drivers who decide to serve customers in the region i. The average earning of drivers R_i is given by:

$$R_i = \rho_i w. \tag{7}$$

Without loss of generality, we assume the wage w is within $[0, 1]$. Similar to some previous literature, we assume the effective labor supply is proportional to the revenue (Hu and Liu, 2023; He et al., 2023). This simplified form of labor supply can well capture the variable marginal cost of recruiting human drivers, which is shown to be a critical characteristic in modeling the difference between HV and AV (Mo et al., 2022) from an aggregate perspective.

Wardrop equilibrium. HV drivers will consider their own utility and switch between the regions until yielding a Wardrop equilibrium as follows:

$$S_i^h(R_i - R_h) = 0, \tag{8}$$

$$R_i \le R_h, \tag{9}$$

$$\sum S_i^h = \bar{S}^h G(R_h), \tag{10}$$

$$0 \le R_h \le 1. \tag{11}$$

Where $G(\cdot)$ is an increasing function. With the assumption that drivers' reservation rates are uniformly distributed over $[0, 1]$, we shall have $G(x) = x, x \in [0, 1]$. Here R_h represents the actual earnings of HV drivers in all regions. When the system achieves a Wardrop equilibrium, there may exist two cases: (1) Interior equilibrium with $R_i = R_h$ for any $i \in N$. In this case, human drivers will provide service in all regions, and the average earnings in these regions are the same. (2) All-or-nothing assignment where human drivers only serve in some regions. We mainly focus on the interior equilibrium situation and the all-or-nothing case is discussed in Appendix.

Platform and social utility. Considering the income from effective customers, the HVs' earnings, and the cost of AVs, the platform maximizes its profit Π_p as follows:

$$\max_{p,w,\gamma_i,S^a} \Pi_p = \sum_i D_i p - \sum_i R_i S_i^h - \sum_i S_i^a \rho_i c_v - S^a c_f. \tag{12}$$

Under the assumption that the numbers of customers and servers are relatively large in a hybrid market, a customer chooses to ride AV or HV based only on the proportion of AV and HV in his/her region, so the amount of in-service AV can be formulated as $\sum_i S_i^a \rho_i$. The welfare of the human drivers Π_h and the customer surplus Π_{cs} are expressed as:

$$\Pi_h = \int_0^{R_h} (R_h - x)\bar{S}^h dx = \frac{\bar{S}^h R_h^2}{2}, \tag{13}$$

$$\Pi_{cs} = \sum_i \int_{p+hW_i}^{1} (x - p - hW_i)\bar{D}_i dx = \frac{1}{2}\sum_i \bar{D}_i (1 - p - hW_i)^2. \tag{14}$$

The social welfare thus can be calculated as $SW = \Pi_p + \Pi_h + \Pi_{cs}$. We summarize the main notations in Table 1.

4. Analysis of market structures

We start the analysis by investigating different market structures. Throughout this section, we investigate a monopoly platform maximizing its profit by Eq. (12). For ease of exposition, we use S_i to represent the total number of vehicles in region i such that $S_i = S_i^h + S_i^a$.

Lemma 1. *In region i, the effective demand D_i increases in the total number of vehicles S_i and converges to $\bar{D}_i(1 - p)$.*

The relation between the effective demand D_i and the vehicles S_i is intuitive. The employment of more vehicles could reduce the waiting time and induce more customers to require service from the platform, whereas the market size and the charged price limit this demand growth. Next, we solve the platform's optimization problem by considering three market states: (1) Pure HV market. The decision variables $S^a = \gamma_i = 0$ and the platform does not provide an AV fleet. (2) Pure AV market. All ride-sourcing vehicles are AVs, and no human driver participates in the market. (3) Hybrid market. Both two types of vehicles provide service in the market.

Table 1
Summary of main notations.

Notation	Definition
$\bar{D}_i$	Potential demand in region i
$\bar{D}$	Sum of potential demand in all regions, calculated by $\sum \bar{D}_i$
D_i	Effective demand in region i
$\bar{S}^h$	Potential supply of human drivers
S_i^h	Effective supply of human drivers in region i
p	Price charged for customers
w	Wage paid to human drivers
R_h	Actual earning of human drivers
S^a	Total number of AVs
S_i^a	Number of AVs in region i
γ_i	Ratio of AVs in region i
c_f	Fixed maintenance cost of one AV
c_v	Variable maintenance cost of one in-service AV
$\bar{c}$	Total maintenance cost of one in-service AV, calculated by $(c_f + c_v)$
S_i	Sum number of AVs and HVs in region i, calculated by $(S_i^h + S_i^a)$
h	Delay sensitivity
ρ_i	Ratio between effective demand and supply in region i
W_i	Expected waiting time in region i
Π_p	Platform profit
Π_h	Labor welfare
Π_{cs}	Customer surplus
SW	Social welfare

Table 2
Effects of the exogenous variables on the equilibrium under pure HV market.

	h	$\bar{D}_i$	$\bar{S}^h$
p^*	↓	↑	↑↓
ρ^*	↓	↑	↑↓
w^*	↑	↑	↓
Π_p^*	↓	N/A	↑

↑(increasing); ↓ (decreasing); ↑↓(increasing and then decreasing).

4.1. Pure HV market

This situation represents a traditional labor market where the platform only employs human drivers. The platform's optimization problem becomes:

$$\max_{p,w} \Pi_p = \sum_i D_i p - \sum_i R_i S_i^h. \tag{15}$$

In the pure HV market, the platform sets the charged price for the customers and the wage provided for the drivers. We have the following findings in the platform's optimal decision (p^*, w^*).

Proposition 1. *In a pure HV market with time-sensitive customers, there exists a unique equilibrium in the price and the wage (p^*, w^*) with $p^* \geq \frac{1}{2}$ and $w^* > 0$ if the labor pool size is sufficiently large ($\bar{S}^h \geq 4h$). The effects of the delay sensitivity h, the potential demand $\bar{D}_i$, and the labor pool size $\bar{S}^h$ on the equilibrium are summarized in Table 2.*

Proposition 1 shows the impacts of market conditions on the platform's optimal decision. When customers lose patience with waiting (increasing h), the platform could directly reduce the price to promote effective demand or recruit more drivers to reduce the waiting time. These measures decrease the platform's revenue, while the human drivers benefit from the employment expansion by receiving improved wages. Meanwhile, the server's utilization rate decreases as too many drivers pour into the platform. In terms of the number of potential customers $\bar{D}_i$, its increase represents a higher arrival rate of customers, so the platform raises the wage to employ more human drivers and boosts the price to maximize profit. Interestingly, the platform's profit does not monotonically increase with the potential demand in a single region, which we describe specifically in the following Corollary 1. Regarding the potential drivers' amount $\bar{S}^h$, it is inversely correlated with the wage. Meanwhile, sufficient drivers can effectively reduce the waiting time and attract more customers, causing an increase in the platform's profit. Remarkably, the increase in the scale of HV drivers cannot always increase the service price. When the labor size is small (the waiting time is large), increasing labor size attracts many customers even if the platform raises the price. When the size reaches a high level, the platform largely reduces the wage level as the competition between laborers intensifies. Without sacrificing net profit $(p - w)$, the platform reduces the price to attract more customers.

Corollary 1. *The optimal revenue of the platform Π_p^* shows a non-monotonic relation to the potential demand in a region $\bar{D}_i$. When all regions have the same demand, Π_p^* increases in $\bar{D}_i$.*

When the demand increases in any region, the platform charges higher prices for the customers and pays higher wages for the drivers. Since the platform charges the same price and pays the same wage in different regions, a change in one region's demand influences the platform's revenue in this region as well as the other region. It is found that, only when the customers equally distribute in all regions, an increase in the demand can always improve the platform's profit.

4.2. Pure AV market

This scenario represents some commercial companies dedicated to self-driving services, like Waymo. These companies do not participate in the labor market; their service strategy is influenced by the development of autonomous vehicles, which is captured by the maintenance cost in our model. The platform's optimization problem becomes:

$$\max_{p,S^a,\gamma_i} \Pi_p = \sum_i D_i(p - c_v) - S^a c_f. \tag{16}$$

In the pure AV market, the platform separately assigns its AV fleet to different regions, making it more adaptive to distinct demands in different regions. To exclude trivial results, we consider the situation where the autonomous vehicle can create profit by providing service such that $p \geq c_v + c_f$. We have the following main findings in the platform's optimal decision $(p^*, S^{a*}, \gamma_i^*)$.

Proposition 2. *In a pure AV market with time-sensitive customers, the platform's optimal decision $(p^*, S^{a*}, \gamma_i^*)$ depends on the maintenance costs (c_f, c_v) and the time sensitivity h.*

(i) The platform could generate profit if the fixed cost c_f is sufficiently low, i.e., $c_f \leq \hat{c_f}(\bar{D}_i, h)$. The threshold $\hat{c_f}(\bar{D}_i, h)$ increases in the potential demand $\bar{D}_i$ and decreases in the time sensitivity h.

(ii) Given $c_f \leq \hat{c_f}(\bar{D}_i, h)$, the charge price p^* increases in the variable cost c_v.

Proposition 2 suggests that a high maintenance cost will prevent the platform from introducing any AV. In particular, as either the potential demand decreases or the customers become more impatient, introducing an AV fleet becomes more difficult for the platform to make profits. Part (ii) shows that the platform will raise its charged price if the variable cost c_v for running an AV becomes higher.

4.3. Hybrid market

The hybrid pattern of HVs and AVs reflects the trend of the future ride-sourcing market. The platforms establish their AV fleets as well as hire human labor. The decision process becomes complex for the platform as it must decide the optimal number of vehicles and adjust the ratio between AVs and HVs. We have the following findings after comparing the hybrid market with the pure HV market.

Proposition 3. *After the platform introduces AVs into ride-sourcing markets, it reduces the wage w^* paid to human drivers. The change of charged price p^* depends on the delay sensitivity h, the potential labor number $\bar{S}^h$, and the maintenance cost (c_f, c_v).*

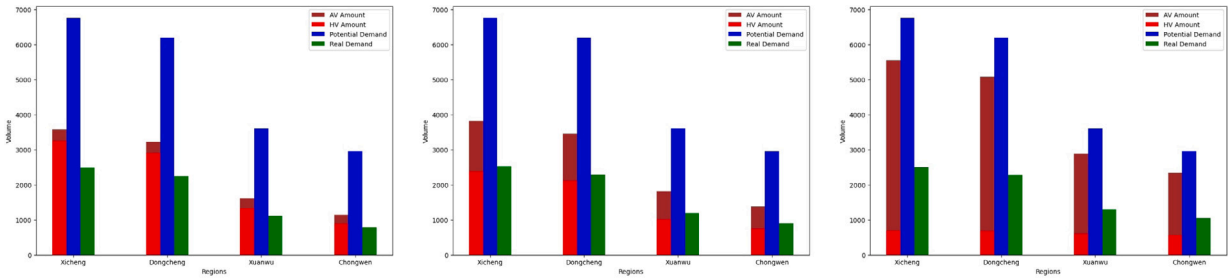
Since the AV fleet serves as an alternative choice in the market, the platform reduces the wage paid to human drivers and deploys some AVs to provide service. Interestingly, introducing the AV fleet does not always reduce the price. In particular, it is observed that the equilibrium price decreases when the delay sensitivity h is smaller than a threshold $\hat{h}(\bar{S}^h, c_v, c_f)$. The threshold $\hat{h}$ increases with $\bar{S}^h$ and decreases with c_v and c_f. The introduction of an AV fleet can reduce the generalized service cost, so the platform intends to stimulate effective demand. The underlying mechanism is that the platform employs two instruments to stimulate the demand, i.e., reducing the price p or the generalized waiting cost hW_i (see Eq. (1)). When customers are not sensitive about waiting (h is small), reducing the price can attract more customers. In contrast, when customers are sensitive about waiting, reducing the generalized waiting cost can attract more customers, and the platform increases the price to improve revenue.

4.4. Case study

We construct a case study to validate the main observations in theoretical modeling and further supplement some additional findings. The dataset is collected in four regions within Beijing Second Ring Road, which contains the customers' demand, drivers' distribution, fee, etc.[6] Based on this dataset, which targets only the pure HV market, we investigate how the introduction of AVs can potentially influence the ride-sourcing market states. Appendix B details how the parameters are chosen and numerical results are calculated. First, we examine the relationship of supply and demand in a hybrid market in Fig. 1.

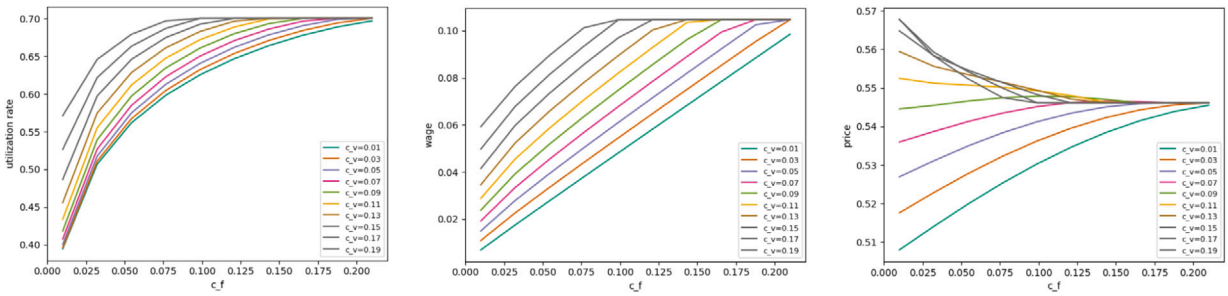
The overall tendency indicates that the platform tends to deploy more AVs when their maintenance costs decrease. The reduction of fixed cost c_f is more profound in promoting the AV market. This is because the platform can deploy more idle AVs to reduce the

[6] Source: https://tianchi.aliyun.com (accessed in November 2023).



(a) $c_f = 0.2,\ c_v = 0$ (b) $c_f = 0.1,\ c_v = 0.1$ (c) $c_f = 0,\ c_v = 0.2$

Fig. 1. Supply and demand states in a hybrid market.



(a) Utilization rate (b) Wage (c) Price

Fig. 2. The effects of AVs on utilization rate, wage, and price.

waiting time for customers, which effectively promotes the demand. Next, we show how the introduction of AVs influences labor welfare and customer surplus in Fig. 2.

The influence on the utilization rate echoes the previous observation. The reduction of the maintenance cost allows the platform to deploy more idle AVs, thus decreasing the utilization rate. The observations on wage and price follow the findings in Proposition 3. The introduction of AVs intuitively reduces the wage paid to human drivers; furthermore, the advance of AVs aggravates the loss in labor welfare. The change in price is more complicated. In general, the price decreases with the variable cost c_v. As this compulsory cost decreases, the platform can employ more in-service AVs and serve more customers after reducing the price. In terms of the fixed cost c_f, the situation depends on how the platform assesses the role of AVs. When the variable cost c_v is at low levels, the reduction of c_f intensifies the inclination to serve more customers, which results in a price decline. However, when c_v is at high levels, the deployment of AVs demonstrates great potential in reducing waiting time instead of providing service. Recall the fact that the effective demand is jointly influenced by waiting time and price. So, the reduction of c_f induces the platform to deploy more idle AVs while charging higher prices, which will not dramatically influence the effective demand.[7]

4.5. Analysis summary

The main findings in distinct market structures are as follows. Firstly, we identify the interior Wardrop Equilibrium's existing conditions for different market states and find that the time sensitivity and the potential demand decide the existence of the equilibrium. In particular, a sufficiently small time sensitivity h could guarantee equilibrium under all market states. Secondly, we compare the market with and without AV fleets. In a pure HV market, we get two interesting findings: (1) the price shows a non-monotonic relation to the labor size because of the trade-off between the waiting time and price in attracting the customers; (2) an increase of the potential demand can improve the platform's profit only when the regions equally share the whole demand, which is mainly because the platform charges the same price and pays the same wage in all regions. In terms of the AV fleet, we point out that the platform should introduce the AV fleet when the maintenance cost is sufficiently low; namely, introducing mature autonomous technology benefits the platform. Regarding the hybrid market, we find that the introduction of AVs induces the platform to reduce the wage paid to human drivers, while the change of charged price depends.

[7] In the appendix, we have shown other numerical results in Fig. B.2, like the platform can increase the revenue when AV technology grows more mature.

Table 3
Equilibrium outcomes under different market structures and optimization objectives.

Optimization Objective	Monopoly Optimum		Social Optimum	
Market Structure	Pure HV	Hybrid	Pure HV	Hybrid
Wage	$\frac{\bar{D}}{2(\bar{D}+\bar{S}^h)}$	$\frac{\bar{c}}{2}$	$\frac{\bar{D}}{\bar{D}+\bar{S}^h}$	$\bar{c}$
Price	$\frac{2\bar{D}+\bar{S}^h}{2(\bar{D}+\bar{S}^h)}$	$\frac{1+\bar{c}}{2}$	$\frac{\bar{D}}{\bar{D}+\bar{S}^h}$	$\bar{c}$
Number of AVs		$\frac{\bar{D}-\bar{c}\bar{D}-\bar{c}\bar{S}^h}{2}$		$\bar{D}-\bar{c}\bar{D}-\bar{c}\bar{S}^h$
Platform Profit	$\frac{\bar{D}\bar{S}^h}{4(\bar{D}+\bar{S}^h)}$	$\frac{1}{4}(\bar{D}(1-\bar{c})^2+\bar{S}^h\bar{c}^2)$	0	0
Labor Welfare	$\frac{\bar{D}^2\bar{S}^h}{8(\bar{D}+\bar{S}^h)^2}$	$\frac{\bar{c}^2\bar{S}^h}{8}$	$\frac{\bar{D}^2\bar{S}^h}{2(\bar{D}+\bar{S}^h)^2}$	$\frac{\bar{c}^2\bar{S}^h}{2}$
Customer Surplus	$\frac{\bar{D}(\bar{S}^h)^2}{8(\bar{D}+\bar{S}^h)^2}$	$\frac{1}{8}(1-\bar{c})^2\bar{D}$	$\frac{\bar{D}(\bar{S}^h)^2}{2(\bar{D}+\bar{S}^h)^2}$	$\frac{1}{2}(1-\bar{c})^2\bar{D}$
Social Welfare	$\frac{3\bar{D}\bar{S}^h}{8(\bar{D}+\bar{S}^h)}$	$\frac{3}{8}(\bar{D}(1-\bar{c})^2+\bar{S}^h\bar{c}^2)$	$\frac{\bar{D}\bar{S}^h}{2(\bar{D}+\bar{S}^h)}$	$\frac{1}{2}(\bar{D}(1-\bar{c})^2+\bar{S}^h\bar{c}^2)$

5. Influences of AVs and relevant regulations

We already have a general picture of how the platform utilizes the AV fleet in the emerging hybrid market. However, due to the calculation complexity, we cannot derive the closed forms of the equilibrium results. Further investigations on the labor market, social welfare, and policy implications cannot be finished with the basic model setting. To complete our analysis and finish the above investigations, we simplify the waiting process ($h = 0$) for tractability. There are two remarks regarding this simplification. First, we follow the common practice in the literature (Li et al., 2022; Benjaafar et al., 2022; Siddiq and Taylor, 2022; Asadpour et al., 2023) to make this assumption, which can help make the analytical derivations more tractable. As we have shown in the previous section, the complexity form of the basic model disables us to derive the solution in closed form, which further limits our comparisons between different optimum cases. In this section, by simplifying the waiting process, we can focus on comparing different optimum situations and further examine the relevant regulations. Second, the primary goal of our study is to investigate how the introduction of AVs affects the market equilibrium in ride-sourcing systems. We have identified the heterogeneous marginal costs of HV and AV as the key drivers resulting in our main conclusions. The main insights highlight the effects of the supply side on the platform and society, and do not depend on the specific demand form. The detailed modeling of the waiting process can bring the model closer to reality; however, the main insights obtained from our model still hold.

This section proceeds as follows. In the first subsection, we examine the influences of AVs by comparing the forthcoming hybrid market with the present pure HV market. The comparison helps us identify AVs' potential impacts on the labor market and social welfare. In the second subsection, we change the optimization objective from the platform's profit (i.e., Eq. (12)) to social welfare. We focus on whether the advance of AVs can benefit society rather than solely improve the platform's profit. Furthermore, we examine which factors can widen or fill the welfare gap between the monopoly and social optimum. In the last subsection, from policymakers' perspectives, we examine the platform's responses to different regulations and identify the regulations that can potentially improve social welfare. For ease of exposition, we define two new notations $\bar{D} \equiv \sum_i \bar{D}_i$ and $\bar{c} \equiv c_v + c_f$. The former represents the total potential demands, and the latter represents the maintenance cost of an in-service AV. In this Section, the equilibrium results are recorded as $(p_p^*, w_p^*, \gamma_{i,p}^*, S_p^{a*})$.

5.1. Pure HV and hybrid markets

In this section, we compare the pure HV market with the hybrid market, representing a foreseeable transformation in ride-sourcing markets.

Lemma 2. *Given $\bar{c} < \frac{\bar{D}}{\bar{D}+\bar{S}^h}$, a monopoly platform pays a lower wage to the human drivers and charges a lower price for the customers after introducing AVs. The details are shown in Table 3.*

Only when the maintenance cost of an in-service AV is sufficiently low (i.e., $\bar{c} < \frac{\bar{D}}{\bar{D}+\bar{S}^h}$), the AV fleet will be introduced by the platform. Generally, the introduction of AVs benefits the platform and the customers, but it reduces labor welfare as the platform reduces the wage paid for human drivers. The competition relationship between HVs and AVs is attributed to the marginal costs. The marginal cost of providing an additional AV is constant, while the marginal cost of recruiting an additional human driver increases with the number of recruited human drivers. The platform can replace some HVs with AVs once the latter requires a sufficiently small maintenance cost. A few interesting observations can be found by comparing the pure HV market with the hybrid market under the situation of monopoly optimum.

First, it is noted that the wage is only influenced by the maintenance cost in the hybrid market. This contrasts with the traditional HV market, where the populations of potential laborers and passengers will directly influence the determined price and wage. When the hybrid market reaches equilibrium, the platform has to allocate its budget to recruiting an AV or a human driver, so the wage paid to the human driver is associated with the AV's marginal cost. The demand does not influence the wage because the platform would recruit more AVs to satisfy the increasing demand. The price also only depends on the AV's condition since the total supply (i.e., $w_p^*\bar{S}^h + S_p^{a*} = \frac{(1-\bar{c})\bar{D}}{2}$) is not influenced by the potential labor size. In other words, once the efficient AV fleet is introduced,

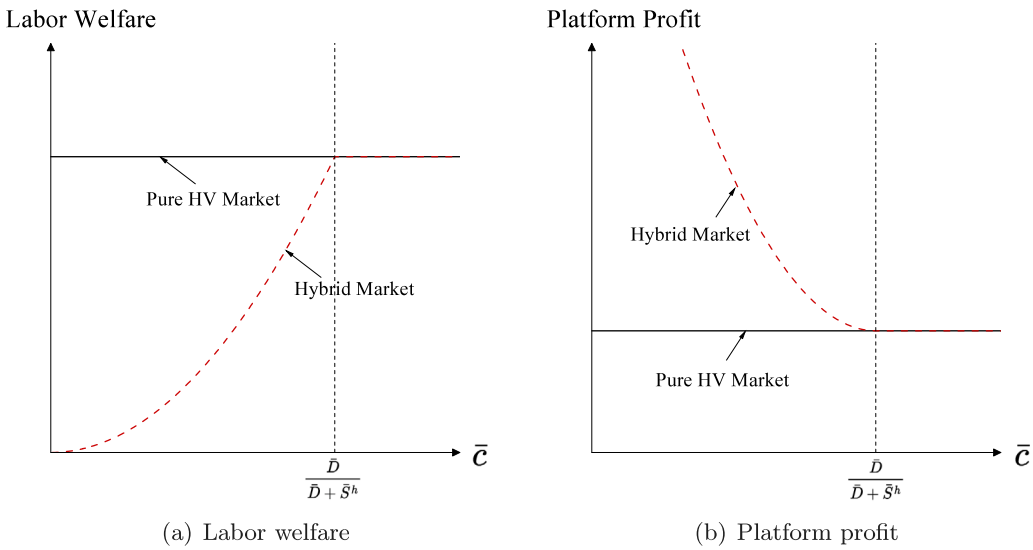


Fig. 3. The influences of AV on labor welfare and platform profit ($\bar{S}^h = 10, \bar{D} = 50$).

it will play a leading role in providing services in the market. Second, in a hybrid market, the proportion of AVs assigned to any region is a special decision variable (i.e., $\gamma^*_{i,p}$). It is not unique but has one constraint under the equilibrium:

$$\gamma^*_{i,p} S^{a*}_p \leq \bar{D}_i(1 - p^*_p). \tag{17}$$

This relation ensures that the number of human drivers is positive.

Third, we can compare the numbers of AVs and HVs in a hybrid market. When $\frac{\bar{D}}{\bar{D}+\bar{S}^h+1} < \bar{c} < \frac{\bar{D}}{\bar{D}+\bar{S}^h}$, the platform employs more HVs than AVs (i.e., $\frac{\bar{c}}{2} > \frac{\bar{D}-\bar{c}\bar{D}-\bar{c}\bar{S}^h}{2}$). In contrast, the platform relies more on AVs in providing service when the technology of AV is mature enough $\bar{c} < \frac{\bar{D}}{\bar{D}+\bar{S}^h+1}$. Combined with the above findings, we can conclude that more HVs may provide service in the hybrid market, but the platform only increases the number of AVs in response to the increased demand. After examining how the monopoly platform adjusts its decision, we investigate how the indexes, including platform profit, labor welfare, customer surplus, and social welfare, are changed after the introduction of AVs.

Proposition 4. *Given $\bar{c} < \frac{\bar{D}}{\bar{D}+\bar{S}^h}$, the introduction of AVs reduces the labor welfare, but it improves the platform profit, the customer surplus, and the social welfare. With a decline in $\bar{c}$, the reduction of labor welfare and the improvements of the three indexes are more significant.*

The advance of AVs can benefit the platform and society, although it might worsen labor welfare. In a hybrid market, the AV fleet is leading in providing services and inevitably squeezes the labor's welfare. This influence becomes more evident as the AV becomes more cost-effective (see Fig. 3(a)) since the platform tends to replace HVs with AVs for providing service. With a decline in $\bar{c}$, the platform also reduces the generalized cost of providing ride-sourcing service. As such, the platform (see Fig. 3(b)), customers, and society benefit from this technological advancement. Nevertheless, it should be noted that the above observations are captured only when the AV technology is mature enough (i.e., $\bar{c} < \frac{\bar{D}}{\bar{D}+\bar{S}^h}$). Otherwise, the platform capable of deploying the AV fleet still operates a pure HV market.

5.2. Monopoly optimum and social optimum

This section examines the welfare gap between two different optimum cases. Monopoly optimum refers to the situation in which a monopoly maximizes its own profit, while social optimum also considers labor welfare and consumer surplus. When investigating the hybrid market, our focus is on the situation that the AV technology is mature enough (i.e., $\bar{c} < \frac{\bar{D}}{\bar{D}+\bar{S}^h}$). In the following proposition, we summarize how the platform adjusts its decisions when the optimization objective changes.

Proposition 5. *After switching from monopoly optimum to social optimum, the platform pays a higher wage to the human drivers, charges a lower price for the customers, and deploys a larger size of AVs. The details are shown in Table 3.*

When maximizing social welfare, the platform should reduce the price and serve as many customers as possible. In particular, the platform sets the price to the same as the wage. By this means, the customer has access to the cheapest service, and the human drivers earn the highest wages. An interesting observation is in the hybrid market. When the platform maximizes its own profit,

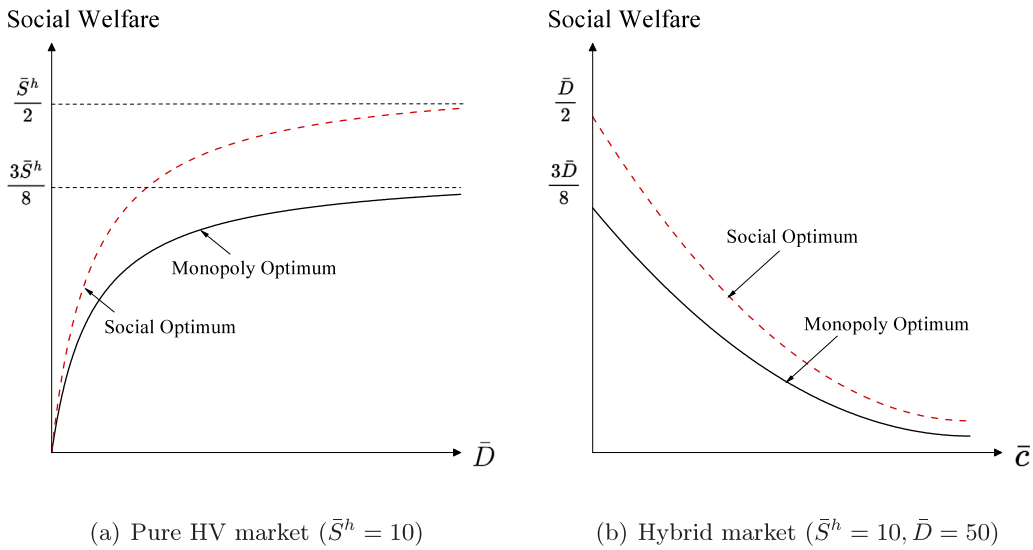


Fig. 4. The social welfare gap between monopoly and social optimums.

it tends to deploy a small size of AV fleet. In contrast, in order to serve more customers under social optimum, the platform still increases the number of AVs, although its own profit cannot be improved. In the following corollary, we analyze how the components of social welfare change with the switch of optimization objectives.

Corollary 2. *After switching from monopoly optimum to social optimum, the labor welfare and customer surplus increase; the platform profit decreases to zero.*

Consistent with our analysis, the labor welfare and customer surplus increase due to the increased wage and decreased price. In a pure HV market, the wage equals the price. In a hybrid market, the platform sets the wage and the price to the same as the maintenance cost. In this light, the platform, under either condition, cannot generate any profit. Next, we compare the social welfare under monopoly optimum with the one under social optimum to identify which factors can widen or fill the gap to the first-best condition.

Corollary 3. *Developing AV technology and stimulating the demand or labor supply can improve social welfare. Nevertheless, the above changes widen the gap in social welfare between the monopoly and social optimums.*

- *In a pure HV market, stimulating the demand or labor supply (i.e., increased $\bar{D}$ or $\bar{S}^h$) widens the welfare gap.*
- *In a hybrid market, except for the demand or labor supply, the advance of AVs (i.e., decreased $\bar{c}$) also widens the welfare gap.*

Generally, the welfare gap to the first-best condition widens because the monopoly platform earns higher profits. An increase in the demand or supply can effectively increase the profit; the former enables the platform to raise the charged price while the latter allows it to reduce the wage paid to the laborers (see Fig. 4(a)). In a hybrid market, the platform replaces some HVs with AVs. The deployment of AVs can reduce the platform's generalized service cost and increase revenue. Therefore, the advance of AVs improves the social welfare under monopoly optimum but widens the welfare gap to the first-best situation (see Fig. 4(b)).

5.3. Regulation outcomes and analysis

Based on previous analysis, a few regulations/policies are discussed in this section to determine how the platform will adjust its decision in response to different regulations and the resulting market equilibrium outcome. Similarly, we focus on a hybrid market with mature AV technology (i.e., $\bar{c} < \frac{\bar{D}}{\bar{D}+\bar{S}^h}$). We can investigate all the potential regulations in the emerging hybrid market through this assumption. For ease of exposition, we record the platform's optimal decision as (w_p^*, p_p^*, S_p^{a*}) under monopoly optimum and (w_s^*, p_s^*, S_s^{a*}) under social optimum. After setting regulations, the platform's revised optimal decision is denoted as ($w_{r,\square}^*$, $p_{r,\square}^*$, $S_{r,\square}^{a*}$). Our analysis proceeds as follows. Firstly, we will examine the platform's responses to distinct regulations, which are already imposed in pure HV markets and potentially adopted in hybrid markets. In particular, by restricting the regulation intensity, we only discuss the regulations that can affect the platform's decisions. Secondly, we sort out the regulations that can effectively improve social welfare, which can provide some managerial insights for policy-makers.

5.3.1. Price-cap regulation

Some governments regulate the price set by the ride-sourcing platforms to prevent them from greedily extracting excessive profit. Under a price-cap regulation, the ride-sourcing platform has to set the price below a regulated price.

Lemma 3. *When a price-cap regulation is imposed ($p < \overline{p}$) and the regulated intensity is $\overline{p} < p_p^*$, the platform holds the wage level ($w_{r,\overline{p}}^* = w_p^*$) and deploys more AVs ($S_{r,\overline{p}}^{a*} > S_p^{a*}$).*

The decline in price means that the platform has to serve more customers. Because the marginal cost of recruiting a human driver increases with the number, the platform would deploy more AVs to satisfy the demand.

5.3.2. Minimum wage

The wage w influences the effective number of human drivers willing to participate in the ride-sourcing market. In some cities like NYC, ride-sourcing drivers are entitled to a minimum wage guarantee. When regulation of minimum wage is set, the wage w paid by the platform to drivers should be larger than $\underline{w}$.

Lemma 4. *When a minimum wage is proposed ($w > \underline{w}$) and the regulated intensity is $\underline{w} > w_p^*$, the platform holds the price ($p_{r,\underline{w}}^* = p_p^*$) and reduces the number of AVs ($S_{r,\underline{w}}^{a*} < S_p^{a*}$).*

More human drivers crowd into the market after a minimum wage standard is proposed. In this light, the platform reduces the size of the AV fleet to prevent the overflowing supply.

5.3.3. Maximum fleet size

Maximum fleet size control is a standard regulation widely adopted in many cities, including Beijing, Shanghai, and NYC. When there is maximum control of the fleet size, the platform has to set the overall fleet size, including AVs and HVs, below a maximum number.

Lemma 5. *When a maximum fleet control is proposed ($S^a + \bar{S}^h w < \overline{S}$) and the regulated intensity is $\overline{S} < S_p^{a*} + \bar{S}^h w_p^*$, the platform holds the wage level ($w_{r,\overline{S}}^* = w_p^*$), increases the charged price ($p_{r,\overline{S}}^* > p_p^*$), and reduces the size of AV fleet ($S_{r,\overline{S}}^{a*} < S_p^{a*}$) .*

The control of total fleet size causes a short-supply market. In response, the platform increases the price and simultaneously reduces the number of AVs to come back in balance.

5.3.4. AV fleet size control

When AVs account for a non-negligible proportion of the ride-sourcing markets, the government may impose some regulations on the AV fleet size. There may be two types of AV fleet size regulatory schemes: (1) the government sets a minimum AV fleet, and the platform is required to offer an AV fleet larger than the minimum AV fleet (minimum AV fleet control, $S^a > \underline{S}^a$); (2) the government sets a maximum AV fleet, and the platform should promise its AV fleet is smaller than the maximum AV fleet (maximum AV fleet control, $S^a < \overline{S}^a$).

Lemma 6. *There are two situations regarding the control of AV fleet size.*

- *When the regulation controls the minimum size ($S^a > \underline{S}^a$) and the control intensity is $\underline{S}^a > S_p^{a*}$, the platform reduces the wage level ($w_{r,\underline{S}^a}^* < w_p^*$) and the charged price ($p_{r,\underline{S}^a}^* < p_p^*$).*
- *When the regulation controls the maximum size ($S^a < \overline{S}^a$) and the control intensity is $\overline{S}^a < S_p^{a*}$, the platform increases the wage level ($w_{r,\overline{S}^a}^* > w_p^*$) and the charged price ($p_{r,\overline{S}^a}^* > p_p^*$).*

The control of AV fleet size induces the platform to adjust the wage level and the charged price. When the size of the AV fleet is required to exceed a standard, the platform reduces the wage paid for human drivers and relies more on AVs to provide service. Simultaneously, to maintain the supply–demand balance, the platform reduces prices to attract more customers. The situation is reversed when the platform is required to shrink the size of the AV fleet.

5.3.5. Improvement of social welfare

Having examined the platform's responses to different regulations, we sort out the regulations that can potentially improve social welfare.

Proposition 6. *Among different regulation regimes, the regulations of price-cap, minimum wage, and minimum AV fleet size can improve social welfare; in contrast, the regulations of maximum fleet size and AV fleet size always worsen social welfare. Details are shown in Table 4 and the specific conditions to secure improvement in social welfare are shown in the appendix.*

In general, regulating the maximum number of service providers in the market always harms society. Such regulations (i.e., maximum fleet size and AV fleet size) cause the platform to charge higher prices and employ fewer service providers, which significantly reduces the platform's profit and customer surplus. The other three regulations can benefit social welfare under most conditions. We discuss these regulations one by one.

Table 4
Implications of different regulations (↑: increasing; ↓: decreasing).

Regulation Regime	Platform Profit	Labor Welfare	Customer Surplus	Social Welfare
Price-cap		no change	↑	depends
Minimum Wage		↑	no change	depends
Maximum Fleet Size	↓	no change	↓	↓
Minimum AV Fleet Size		↓	↑	depends
Maximum AV Fleet Size		↑	↓	↓

The regulation of price-cap improves customer surplus without influencing labor surplus. The platform deploys more AVs to satisfy the stimulating demand caused by the price decline. This policy reduces social welfare only when (1) there is a shortage in labor market (i.e., $\bar{S}^h < 2\bar{D}$), (2) the AV's maintenance is costly (i.e., $\bar{c} > \frac{1}{3}$), and (3) the price standard is too low (i.e., $\overline{p} < \frac{1}{2}(3\bar{c}-1)$). The price cap effectively improves social welfare when the supply is sufficient, and AV technology is mature enough.

Minimum wage regulation improves labor welfare and does not influence customer surplus. The platform replaces some AVs with HVs in the market. This policy reduces social welfare only when (1) the AV technology is very mature (i.e., $\bar{c} < \min\{\frac{2}{3}, \frac{\bar{D}}{\bar{D}+\bar{S}^h}\}$) and (2) the wage standard is too high (i.e., $\underline{w} > \frac{3\bar{c}}{2}$). In other words, when the erosion of AVs in ride-sourcing markets is inevitable, the government should not insist on guaranteeing labor welfare.

The minimum AV fleet size regulation harms labor welfare but greatly enhances customer surplus. This policy reduces social welfare only when (1) the AV's maintenance is costly (i.e., $\bar{c} > \frac{2\bar{D}}{3\bar{D}+3\bar{S}^h}$) and (2) the fleet size of AVs is required to be too large (i.e., $\underline{S}^a > \frac{3}{2}(\bar{D} - \bar{c}\bar{D} - \bar{c}\bar{S}^h)$). This finding highlights the value of AVs in ride-sourcing markets. Once the technology becomes cost-efficient, customers can enjoy a relatively cheap service, and social welfare can be improved.

5.4. Analysis summary

The major managerial insights in this section are summarized as follows. First, we compare the pure HV market with the hybrid market, representing a foreseeable transformation in ride-sourcing markets. Since the AV requires a constant marginal cost for providing service, the introduction of the AV fleet undoubtedly replaces some HVs in providing service and erodes labor welfare. Nevertheless, as cost-efficient service providers, AVs with sufficiently low maintenance costs can benefit the platform, the customers, and society.

Second, we compare the monopoly optimum with the social optimum and identify the causes of the welfare gap. To maximize social welfare, the monopoly platform is expected to raise wages, charge lower prices, and deploy more AVs. Interestingly, the advance of AV technology improves social welfare but also widens the social welfare gap between the monopoly and social optimums. In other words, because the platform is the one to deploy the AV fleet, the improvement of AV is more beneficial to the platform than to society.

Last, we compare different regulation regimes of governments and sort out the ones that can improve social welfare. Specifically, price cap regulations, minimum wage, and minimum AV fleet size can improve social welfare under most conditions. Nevertheless, the regulation intensity should be moderate. Furthermore, the adoption of regulations should depend on the demand-supply relations and the condition of AV technology. For example, the government can set the minimum wage standard at the initial stages of AV's development. In contrast, the government can regulate the price cap and the minimum AV fleet size when the AV technology is mature enough.

6. Extension: Spatial relocation problem

The introduction of AVs is promising in handling spatial imbalance of transportation networks. Because the spatial process is simplified in the basic setting, we employ this extension to examine a spatial relocation problem in a more generalized form. Different from the centrally controlled AV fleet, human drivers can generate a propensity for relocation after a trip. This explanation can be supported by recent literature in studying how laborers respond to the platform's instructions or recommendations (Liu et al., 2023). In our context, the human drivers do not directly follow the platform's relocation suggestion because they may suspect the dependability of expected earnings for outbound trips. In addition to the labor's aversion towards the platform's suggestion, the irrational behavior can be broadly caused by "status quo bias" (Samuelson and Zeckhauser, 1988). That is, the human drivers stick with serving the local trip even if the outbound trip can lead to a better-expected outcome. From another perspective, because the platform is better informed of the traffic conditions and makes centralized decisions on the relocation of AVs in different regions, human drivers can incur an additional cost for relocation when compared with AVs.

6.1. Local and outbound trips

Following recent literature (Zhang and Nie, 2022) in studying the relocation problem, we differentiate the extension with the basic model (in Section 5) as follows. (1) The potential demand in region i consists of two parts: the trip in this region $\bar{D}_{ii}$ (**local trip**) and the trip from this region to the others $\bar{D}_{ij}$ (**outbound trip**), where $i, j \in N$ and $i \neq j$. Hence, the potential demand in region i can be calculated by $\bar{D}_i = \bar{D}_{ii} + \sum_j \bar{D}_{ij}$. The transfer ratio $t_i \equiv \frac{\sum_j \bar{D}_{ij}}{\bar{D}_i}$ (or $1 - t_i \equiv \frac{\bar{D}_{ii}}{\bar{D}_i}$) denotes the ratio between these two

Table 5
Equilibrium (monopoly optimum) in a pure HV market (with spatial relocation).

Market Structure	Pure HV (local trip only)	Pure HV (both trips)
Optimization Objective	Monopoly Optimum	
Wage	$\frac{\bar{D}(1-\bar{t})}{2(\bar{D}+\bar{S}^h)-2\bar{D}\bar{t}}$	$\frac{(\beta-1)\bar{D}\bar{t}+\bar{D}}{2((\beta-1)\bar{D}\bar{t}+\bar{D}+\beta\bar{S}^h)}$
Price	$1-\frac{\bar{S}^h}{2(\bar{D}+\bar{S}^h)-2\bar{D}\bar{t}}$	$\frac{2\bar{D}((\beta-1)\bar{t}+1)+\beta\bar{S}^h}{2((\beta-1)\bar{D}\bar{t}+\bar{D}+\beta\bar{S}^h)}$
Platform Profit	$\frac{\bar{D}\bar{S}^h(1-\bar{t})}{4(\bar{D}+\bar{S}-\bar{D}\bar{t})}$	$\frac{\beta\bar{D}\bar{S}^h}{4((\beta-1)\bar{D}\bar{t}+\bar{D}+\beta\bar{S}^h)}$
Social Welfare	$\frac{3\bar{D}\bar{S}^h(1-\bar{t})}{8(\bar{D}+\bar{S}-\bar{D}\bar{t})}$	$\frac{\beta\bar{D}\bar{S}^h(\bar{D}((\beta-1)\bar{t}+1)(\beta+(\beta-1)\beta\bar{t}+2)+3\beta\bar{S}^h)}{8((\beta-1)\bar{D}\bar{t}+\bar{D}+\beta\bar{S}^h)^2}$
Utilization Rate	1	$\frac{\beta}{1-(1-\beta)\bar{t}}$
Service Rate	$1-\bar{t}$	1

demands. An increased ratio indicates that the outbound trip accounts for a larger part of the effective demand. (2) When reaching a Wardrop equilibrium (i.e., Eq. (8)), a parameter β is added to the human driver's actual earnings for outbound trips. That is, under the interior equilibrium, the relation $R_{ii} = \beta R_{ij}$ holds. By replacing R with ρw in Eq. (7), the relation changes to $\rho_{ii} = \beta\rho_{ij}$. With a decline in β, the propensity towards relocation aversion becomes more significant and the human drivers are more inclined to serve the local trips.

Given the primary objective is to analyze the impacts of relocation factors (i.e., transfer rate and relocation propensity) on the market equilibrium, the following assumption is made for computationally tractable. For any given region i, it is assumed that the number of outbound trips originating from this region equals the aggregate of outbound trips from other regions directed towards this region, i.e., $\sum_j \bar{D}_{ij} = \sum_j \bar{D}_{ji}$. This assumption enables the model to depict a simplified dynamic balance in the network, with no net flow generated by the outbound trip.

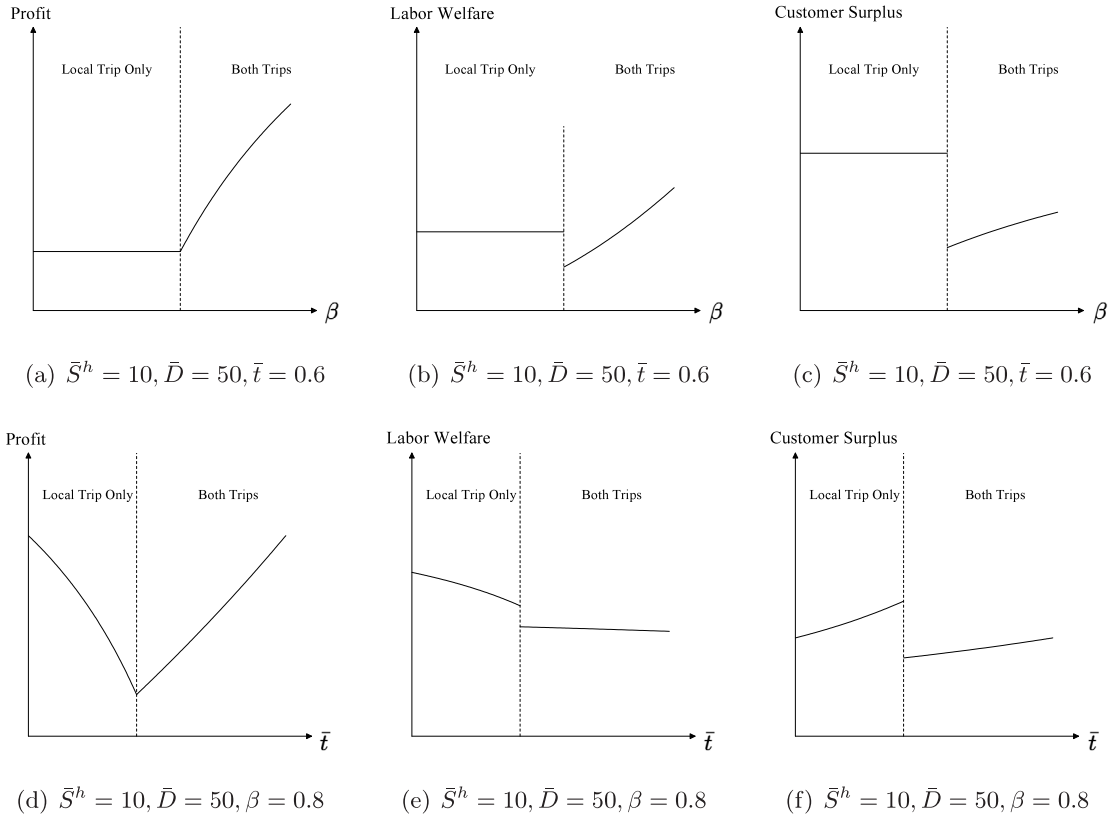
Given the aforementioned assumption and the waiting process is simplified ($h = 0$), we can conduct a preliminary analysis regarding the platform's strategy. When running the pure HV market, there exist two potential optimal cases. (1) The platform employs a small group of human drivers to provide service for only the local trips (i.e., the all-or-nothing case with $\rho_{ii} = 1$). (2) The platform employs a large group of human drivers to provide service for both the local and outbound trips, which causes an oversupply for local trips (i.e., interior equilibrium case with $\rho_{ij} = 1$ and $\rho_{ii} = \beta < 1$). Under either situation, the idle vehicle never exists in the outbound trips and might exist in the local trips. Hence, our model precludes considerations such as compensations for return trips (Rosaia, 2020; Brancaccio et al., 2023). For ease of exposition, we employ the notation $T_{out} \equiv \sum_i \bar{D}_i t_i$ to represent the sum of all potential outbound demand and $T_{in} \equiv \sum_i \bar{D}_i(1-t_i)$ for the potential local demand. We use the notation $\bar{t} \equiv T_{out}/\bar{D}$ to represent the transfer ratio at the aggregate level, where $\bar{D} \equiv \sum_i \bar{D}_i$.

6.2. Pure HV market

The outcomes for a pure HV market at equilibrium are detailed in Table 5 and Fig. 5. The platform transitions from serving local trips to both areas when the propensity becomes relatively weak (i.e., $\beta > \frac{\bar{D}(1-\bar{t})^2}{\bar{D}(1-\bar{t})^2+\bar{S}^h\bar{t}}$) or the overall transfer rate is large enough (i.e., $\bar{t} > \frac{2\beta\bar{D}-2\bar{D}-\beta\bar{S}^h}{2(\beta-1)\bar{D}} - \frac{1}{2}\sqrt{\frac{-4\beta^2\bar{D}\bar{S}^h+4\beta\bar{D}\bar{S}^h+(\beta\bar{S}^h)^2}{(\beta-1)^2\bar{D}^2}}$). At this transition, labor welfare notably decreases as the actual earnings of human drivers drop from w to βw. Serving only local trips results in a utilization rate of 1 for human drivers. However, this rate diminishes when the platform serves both trips. When focusing solely on local trips, the platform sets a lower price to maintain sufficient local demand. However, at the transition point, it raises the price, leading to a decrease in customer surplus.

The effect of propensity β is straightforward. If the propensity is significant (i.e., small β), the platform opts to serve only local trips. In cases where the platform serves local trips, the relocation propensity has no impact on different welfare metrics as all effective demand and supply are concentrated in the local markets. The situation changes when the platform begins to serve both trips. In general, a decline in this propensity (i.e., increased β) benefits the platform, the labor, and the customers. For the platform, as human drivers are more willing to provide service for outbound trips, it can lower the price to boost demand and complete more transactions. For the laborers, as β increases, the actual earning (i.e., βw) under equilibrium also increases, improving labor welfare.

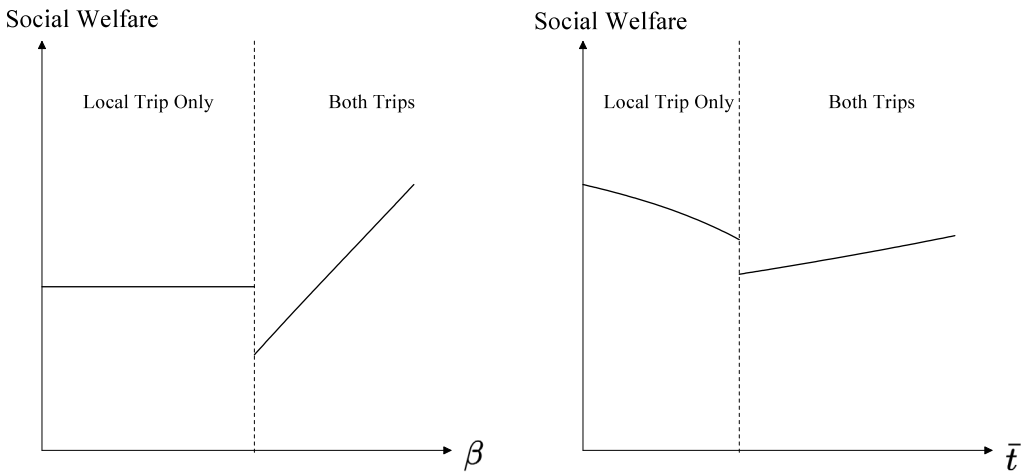
The impact of transfer rate $\bar{t}$ is more complex. An intriguing observation is that the platform's profit is at its lowest when the rate is at a middle level (i.e., the transition point between the two strategies). When the platform serves the local trip only, an increase in the transfer rate implies a decrease in the effective demand. In response, the platform reduces the wage to employ fewer drivers (lowering labor welfare) and lowers the price (raising customer surplus) to secure enough demand in local markets. Nevertheless, an increased transfer rate still results in a loss for the platform. In cases where the platform serves both trips, an increase in the rate is beneficial for the platform and the customers. The underlying mechanism is that the increased transfer rate can counteract the negative effects arising from the relocation propensity. With an increase in the transfer rate, the demand for local trips decreases and the human drivers become more willing to provide service for outbound trips. In this light, the stronger willingness enables the platform to reduce the wage (lowering labor welfare) and correspondingly improves the profit. In particular, when all demands are for the outbound trips (i.e., $t = 1$), no human drivers will gather in the local trips although they exhibit a relocation aversion. At that time, the utilization rate $\frac{\beta}{1-(1-\beta)\bar{t}}$ rises to one and the platform maximizes its profit.



(a) $\bar{S}^h = 10, \bar{D} = 50, \bar{t} = 0.6$ (b) $\bar{S}^h = 10, \bar{D} = 50, \bar{t} = 0.6$ (c) $\bar{S}^h = 10, \bar{D} = 50, \bar{t} = 0.6$

(d) $\bar{S}^h = 10, \bar{D} = 50, \beta = 0.8$ (e) $\bar{S}^h = 10, \bar{D} = 50, \beta = 0.8$ (f) $\bar{S}^h = 10, \bar{D} = 50, \beta = 0.8$

Fig. 5. The impacts of propensity and transfer rate on the platform's profit, labor welfare, and customer surplus.



(a) $\bar{S}^h = 10, \bar{D} = 50, \bar{t} = 0.6$ (b) $\bar{S}^h = 10, \bar{D} = 50, \beta = 0.8$

Fig. 6. The impacts of propensity and transfer rate on social welfare under monopoly optimum.

Combining the above findings, Fig. 6 demonstrates how the propensity β and transfer rate $\bar{t}$ affect social welfare. The propensity has no impact on the welfare when the platform serves the local trip only. However, a reduced propensity benefits the platform, the customers, the labor, and social welfare when serving both trips. The transfer rate helps counter the negative effects of the relocation propensity. Nevertheless, it plays a positive role only when the platform serves both trips. Generally, social welfare first decreases and increases with the rate, reaching a minimum when the platform changes the operation mode.

Table 6
Equilibrium (social optimum) in a pure HV market (with spatial relocation).

Market Structure	Pure HV (local trip only)	Pure HV (both trips)
Optimization Objective	Social Optimum	
Wage	$\frac{\bar{D}(1-\bar{t})}{\bar{D}+\bar{S}^h-\bar{D}\bar{t}}$	$\frac{2\bar{D}(1-\bar{t}+\beta\bar{t})}{2((\beta-1)\bar{D}\bar{t}+\bar{D})+\beta\bar{S}^h}$
Price	$\frac{\bar{D}(1-\bar{t})}{\bar{D}+\bar{S}^h-\bar{D}\bar{t}}$	$\frac{2\bar{D}(1-\bar{t}+\beta\bar{t})}{2((\beta-1)\bar{D}\bar{t}+\bar{D})+\beta\bar{S}^h}$
Platform Profit	0	0
Social Welfare	$\frac{\bar{D}\bar{S}^h(1-\bar{t})}{2(\bar{D}+\bar{S}^h-\bar{D}\bar{t})}$	$\frac{\beta^2\bar{D}\bar{S}^h\left(4\bar{D}((\beta-1)\bar{t}+1)^2+\bar{S}^h\right)}{2(2\bar{D}((\beta-1)\bar{t}+1)+\beta\bar{S}^h)^2}$
Utilization Rate	1	$\frac{\beta}{2\left(1-(1-\beta)\bar{t}\right)}$
Service Rate	$1-\bar{t}$	1

Table 7
Equilibrium (monopoly optimum) in a hybrid market (with spatial relocation).

Market Structure	Hybrid Market (AV-dominant)	Hybrid Market (HV-dominant)
Wage	$\frac{\bar{c}}{2}$	$\frac{(1-\beta)\bar{c}\bar{D}(1-\bar{t})((\beta-1)\bar{t}+1)-2\beta^2\bar{c}\bar{S}^h+\beta\bar{D}(\beta-\beta\bar{t}+\bar{t}-1)}{(1-\beta)^2\bar{D}(1-\bar{t})^2-4\beta^2\bar{S}^h}$
Price	$\frac{1+\bar{c}}{2}$	$\frac{(1-\beta)^2\bar{D}(1-\bar{t})^2-\beta\bar{S}^h(2\beta+\bar{c}(\beta+(\beta-1)\bar{t}+1))}{(1-\beta)^2\bar{D}(1-\bar{t})^2-4\beta^2\bar{S}^h}$
Number of AVs	$\frac{\bar{D}-\bar{c}\bar{D}-\bar{c}\bar{S}^h}{2}$	$\frac{\beta\bar{S}^h(2\beta\bar{c}\bar{S}^h-\bar{D}(\beta-2\bar{c}((\beta-1)\bar{t}+1)+(\beta-1)\bar{t}+1))}{(1-\beta)^2\bar{D}(1-\bar{t})^2-4\beta^2\bar{S}^h}$
Platform Profit	$\frac{1}{4}(\bar{D}(1-\bar{c})^2+\bar{S}^h\bar{c}^2)$	$\frac{\beta\bar{S}^h\left(-\beta\bar{c}^2\bar{S}^h-\beta(1-\bar{c})\bar{D}(1-\bar{c}\bar{t})+(1-\bar{c})\bar{c}\bar{D}(1-\bar{t})\right)}{(1-\beta)^2\bar{D}(1-\bar{t})^2-4\beta^2\bar{S}^h}$
Utilization Rate	1	$\frac{\beta}{1-(1-\beta)\bar{t}}$
Service Rate	1	1

Moving forward, we analyze the difference between social and monopoly optimums. The equilibrium results are collected in Table 6. In line with Proposition 5, the platform reduces the price and increases the wage when prioritizing social welfare. However, due to employing more idle human drivers, the utilization rate decreases when the platform serves both trips.

6.3. Hybrid market

Last, we examine the situation after introducing AV into the market. The AV fleet exhibits an advantage over HVs in serving outbound trips, so there exist two possible situations in a hybrid market. (1) When the maintenance cost for AV is relatively low, the platform entirely relies on AVs to provide service for outbound trips (defined as a **AV-dominant hybrid market**). In other words, conditional on the platform's deployment of AVs, all human drivers choose to provide service for local trips. (2) When the maintenance cost for AV is relatively high, the platform would partially rely on AVs to provide service for outbound trips. In this light, human drivers would provide service for both local and outbound trips (defined as a **HV-dominant hybrid market**). The equilibrium results in a hybrid market are summarized in Table 7.

Similarly, we can compare the platform profit under two situations. The condition to secure the platform's choice of any strategy is not unique. However, the general tendency is that the platform adopts the AV-dominant operation as the maintenance cost decreases. We can find that the AV-dominant hybrid market helps the system to entirely mitigate the negative effect caused by relocation propensity, so the equilibrium results are the same as the ones in Section 5. The system can operate in the most efficient pattern as both the utilization and service rates are maximized. While the HV-dominant hybrid market exhibits identical features to the pure HV one. The primary reason is that the AV fleet just serves some customers in the outbound trips and the platform heavily relies on human drivers to provide service for both trips. As such, the market equilibrium is still driven by the human drivers' biased individual decisions. So in general, the introduction of AV can increase the system efficiency and can mitigate the negative effect caused by relocation propensity. However, this improvement happens only when the technology is mature enough and the AV fleet accounts for a large part of the servers in the market.

7. Conclusions

This paper discusses a mixed market composed of human and autonomous vehicles. Under different market structures, we first identify the existing equilibrium conditions and the monopoly platform's optimal decisions. Some main findings are as follows: (1) In pure HV markets, the price shows a non-monotonic relation to the labor size, and an increase in the potential demand improves the platform's profit only when the demands are equal in different regions. (2) In pure AV markets, the platform should introduce the AV fleet when the autonomous driving technology is mature enough. (3) In hybrid markets, introducing AVs induces the platform to reduce the wage paid to human drivers, while the change of charged price depends.

Furthermore, to figure out tractable results and provide more managerial insights for regulating the mixed fleets, we investigate a hybrid market assuming customers are insensitive about waiting. We compare the pure HV market with the hybrid market. Once the maintenance cost of AVs reduces to a decent level, the penetration of AVs into ride-sourcing markets can benefit the

platform, the customers, and society. Nevertheless, labor welfare suffers a loss due to the inevitable competition. Furthermore, we compare monopoly and social optimums in hybrid markets. To maximize social welfare, the monopoly platform is expected to raise wages, charge lower prices, and deploy more AVs. Interestingly, the advance of AV technology improves social welfare but also widens the social welfare gap between the monopoly and social optimums. Lastly, we compare different regulation regimes of governments. Specifically, the regulations of price cap, minimum wage, and minimum AV fleet size can improve social welfare under most conditions. Nevertheless, policymakers should avoid the extreme intensity of regulations and adopt specific regulations according to demand-supply and AV technology conditions.

Our study opens up several avenues for future extensions. First, we develop an aggregate model that focuses on the change in the aggregate demand and supply in a city after the introduction of AVs, which overlooks the detailed spatiotemporal distributions of demand and supply. It will be interesting to explore the potential of AVs in mitigating spatial supply–demand imbalance among multi-regions (we provide an example in the extension). Second, to make the analytical derivations more tractable, we make some simplifying assumptions (e.g., the price linearly changes waiting time and price; the labor size is proportional to wage). These assumptions limit the generality of the results to some extent, and more sophisticated forms of the demand and supply functions can be considered in future works. Third, multiple queues can exist for different vehicles and priority in matching merits further investigation. Fourth, different regulations under the context of distinct operational patterns (an example is provided in Appendix A) also warrant further examinations; for example, the AV fleet can be owned by the platform or individuals, then the optimal operation and regulation strategies could be different for these two business models.

CRediT authorship contribution statement

Zhenxiao Chen: Methodology, Software, Investigation, Writing – original draft. **Yuhan Miao:** Software, Investigation, Writing – original draft. **Jintao Ke:** Conceptualization, Methodology, Investigation, Funding acquisition, Writing – review & editing. **Qiao-Chu He:** Conceptualization, Methodology, Investigation, Supervision, Writing – review & editing.

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Appendix A. Discussions on competitive fringe

We discuss the problem of competitive fringe in this section. The platform is no longer a monopoly but is rather dominant in the market. The labor supply in the market is constant at N, which is not controlled by the platform. The platform deploys the AV size S^a and decides the price p to maximize its expected utility. The ratio between the number of AVs and the total supply thus can be expressed by $\theta = \frac{S^a}{N+S^a}$. Consistent with Section 5, we simplify the waiting process. The expected utility of the platform can be expressed as:

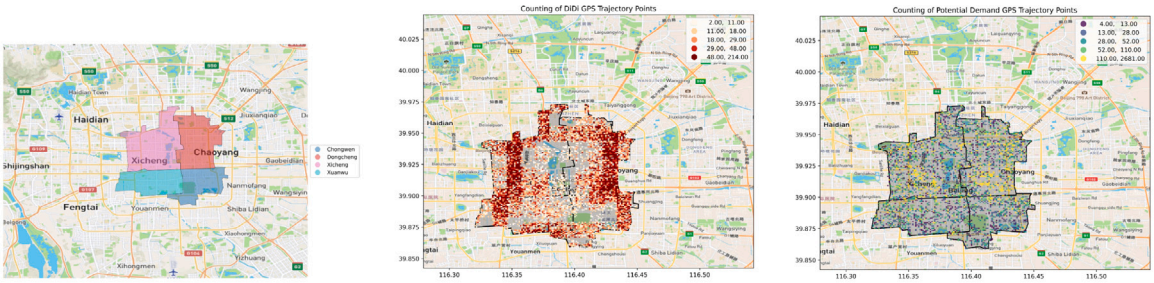
$$\max_{p,S^a} \Pi_p^B = \underbrace{(1-p)\bar{D}\theta(p-c_v)}_{\text{net profit of in-service AVs}} - S^a c_f. \tag{A.1}$$

By solving this optimization problem, we can find the equilibrium results are $p^* = \frac{1+c_v}{2} < \frac{1+\bar{c}}{2}$ and $S^{a*} = \frac{\sqrt{c_f(1-c_v)^2\bar{D}N}}{2c_f} - N$. Compared with the basic setting, we can find that the platform reduces the price p^* after considering competitive fringe; the fleet size of AVs S^{a*} decreases with the number of existing laborers in the market. A supply–demand imbalance is observed. The underlying reason is that the platform cannot generate profit from human drivers, and it has an incentive to employ more AVs to serve more customers in the market.

Appendix B. Case study description

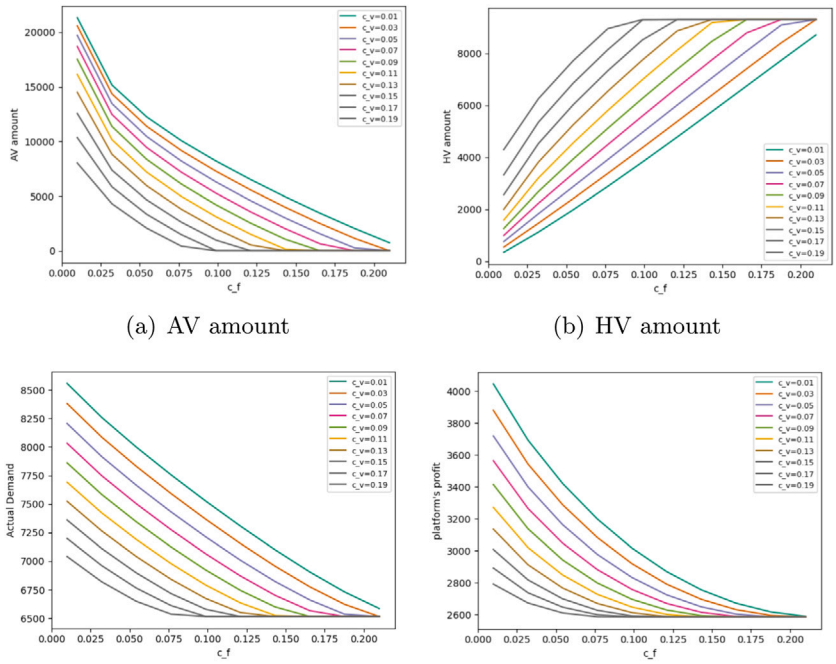
We conduct a numerical experiment using an extensive dataset provided by an integrated company DiDiKuaiDi, a comprehensive company occupying most of the ride-hailing services in China. The dataset contains customers' demand, drivers' distribution, response time, fee, satisfaction level, and hours and locations during the service in Beijing in a week during 2016 (Fig. B.1(a)). Four regions within Beijing Second Ring Road were selected because the geographic conditions and populations are similar. Also, back in 2016, there were no auto-driving vehicles for commercial usage, so we suppose there are no AVs in the market, and the regions are the same except for the potential demand for ride-hailing services. With the location information of both drivers' distribution and customers' demand, we may track and collect the trajectory of supply and potential demand to acquire a precise estimation of $\bar{D}_i$ and S_i^h. As can be seen in the plot figure, most of the drivers' trajectories are distributed among the main roads (Fig. B.1(b)), while the demand is sporadically located in the residential areas (Fig. B.1(c)). The total amount of served customers is calculated by a weighted average of transaction history, as the utilization rate of every region is the same in our stylized settings.

We set up the initial value of HV drivers' wage as $w = \frac{1}{10}$, which can be standardized to any value within $(0, 1)$. A smaller initial value can mitigate the constraint of total driver capacity, which is irrelevant to our study. We assume that the actual price setting



(a) Selected regions (b) Driver distribution (c) Demand distribution

Fig. B.1. The demand and supply states.



(a) AV amount (b) HV amount

(c) Effective demand (d) Platform profit

Fig. B.2. The effects of AVs on vehicle amount, effective demand, and platform profit.

is already optimal in a pure HV market so that the comparison between the experimental outcome and the reality is meaningful, and it will be feasible to learn the actual price $\hat{p}_0$ and customers' time sensitivity $\hat{h}$ in an inverse optimization scheme. The actual price $\hat{p}_0 = 0.545$, and the coefficient of time sensitivity $\hat{h} = 80.504$. The optimization problem can be integrated as:

$$\max_{p,w,S_a^i} \sum_{i=1}^{4} [p\rho(S_i^h + S_i^a) - wS_i^h - c_v\rho S_i^a - c_f S_i^a], \tag{B.1}$$

$$s.t.\ \rho(S_i^h + S_i^a) = \bar{D}_i(1 - p - h\frac{\rho}{(1-\rho)S_i}), \forall i \in \{1,2,3,4\}, \tag{B.2}$$

$$\sum_{i=1}^{4} S_i^h = \bar{S}^h w\rho, \tag{B.3}$$

$$p, w < 1. \tag{B.4}$$

Appendix C. Proofs

We provide detailed proof of the main results. For simplification, we mark:

$$S_i = S_i^a + S_i^h,$$
$$B_{(i)} = S_{(i)} + \bar{D}_{(i)}(1-p) + \frac{\bar{D}_{(i)}h}{S_{(i)}},$$
$$C_{(i)} = \bar{D}_{(i)}(1-p)S_{(i)}.$$

Proof of Lemma 1. Solving

$$\frac{\partial D_i}{\partial S_i} = \frac{\frac{\partial B_i}{\partial S_i}(\sqrt{B_i^2 - 4C_i} - B_i) + 2\frac{\partial C_i}{\partial S_i}}{2\sqrt{B_i^2 - 4C_i}} = 0, \tag{C.1}$$

we can get:

$$S_i = \frac{\bar{D}_i(1-p) \pm \sqrt{\bar{D}_i^2(1-p)^2 + 16\bar{D}_i h}}{4}. \tag{C.2}$$

The smaller solution turns out to be negative while the larger one is extraneous due to the square root, so $\frac{\partial D_i}{\partial S_i} > 0$ for all $S_i > D_i$. Moreover,

$$\begin{aligned} &\lim_{S_i \to \infty} D_i = \bar{D}_i(1-p), \\ &\lim_{S_i \to \infty} \frac{\partial D_i}{\partial S_i} = O(S_i^{-3}), \end{aligned} \tag{C.3}$$

indicates that D_i rapidly converges to $\bar{D}_i(1-p)$. □

Proof of Proposition 1. We finish the proof in two steps. In the following statements, we first present the prerequisites where interior equilibrium exists under the HV market. The interior Wardrop equilibrium (Eq. (C.4) to Eq. (C.5)) can be rewritten as below:

$$\sum_i S_i = \bar{S}^h w \rho_i, \ \forall i, \tag{C.4}$$

$$\rho_i = \rho, \ \forall i. \tag{C.5}$$

Eq. (C.4) implies

$$S_i = \frac{\pm\sqrt{\bar{D}_i^2(1-p)^2 - \frac{4\bar{D}_i h\rho^2}{1-\rho}} + \bar{D}_i(1-p)}{2\rho}, \ \forall i, \tag{C.6}$$

and with $\rho = \frac{\sum_i S_i}{\bar{S}^h w}$ we can get

$$f(\rho) = 2\rho^2\bar{S}^h w - (\sum_i \bar{D}_i(1-p) \mp \sum_i \sqrt{\bar{D}_i^2(1-p)^2 - \frac{4\bar{D}_i h\rho^2}{1-\rho}}) = 0, \tag{C.7}$$

$$\frac{\partial f(\rho)}{\partial \rho} = 4\rho\bar{S}^h w \pm \sum_i \frac{4\bar{D}_i h\rho - 2\bar{D}_i h\rho^2}{(1-\rho)^2\sqrt{\bar{D}_i^2(1-p)^2 - \frac{4\bar{D}_i h\rho^2}{1-\rho}}}. \tag{C.8}$$

Before further exploration, as ρ_{max} makes

$$\min_i \bar{D}_i^2(1-p)^2 - \frac{4\bar{D}_i h\rho_{max}^2}{1-\rho_{max}} = 0, \tag{C.9}$$

we assume $\bar{D}_1$ is the smallest among $\bar{D}_i$. Then

$$\rho_{max} = \frac{\sqrt{\bar{D}_1^2(1-p)^4 + 16\bar{D}_1 h(1-p)^2} - \bar{D}_1(1-p)^2}{8h}. \tag{C.10}$$

1. While $\pm$ is regarded positive, it is obvious that $\frac{\partial f_1(\rho)}{\partial \rho} \geq 0$ and $f_1(0) < 0$, so there might exist another ρ_1 such that $f_1(\rho_1) = 0$, iff $f_1(\rho_{max}) \geq 0$. In such circumstances,

$$\begin{aligned} f_1(\rho_{max}) =& \bar{S}^h w\bar{D}_1(1-p)^2 \frac{\bar{D}_1(1-p)^2 + 8h - \sqrt{\bar{D}_1^2(1-p)^4 + 16\bar{D}_1 h(1-p)^2}}{16h^2} \\ &- (\sum_i \bar{D}_i)(1-p) - \sqrt{\min_{j\neq 1} \bar{D}_j^2 - \bar{D}_1\bar{D}_j}(1-p) \geq 0. \end{aligned} \tag{C.11}$$

2. While $\pm$ is regarded negative, $\frac{\partial f_{i\neq 1}(\rho)}{\partial \rho}(0) = 0$. We set

$$g(\rho) = 4\bar{S}^h w - \sum_i \frac{4\bar{D}_1 h - 2\bar{D}_1 h\rho}{(1-\rho)^2\sqrt{\bar{D}_1^2(1-p)^2 - \frac{4\bar{D}_1 h\rho^2}{1-\rho}}}. \tag{C.12}$$

Notice that $g(0) = 4\bar{S}^h w - \frac{8h}{1-p}$ and that

$$\frac{\partial g(\rho)}{\partial \rho} = -\sum_{i=1}^{2} \frac{4\bar{D}_i h\rho - 2\bar{D}_i h\rho^2 + 2\bar{D}_i h(\rho-1)(\rho-3)(D_i^2(1-p)^2 - \frac{4D_i h}{\rho}^2 1-\rho)}{(1-\rho)^4(\bar{D}_i^2(1-p)^2 - \frac{4\bar{D}_i h\rho^2}{1-\rho})^{\frac{3}{2}}} < 0, \tag{C.13}$$

so if $g(0) > 0$, i.e. $\bar{S}^h w > \frac{2h}{1-p}$, then there exists another $\rho_2 > 0$ such that $f_j(\rho_2) = 0$, iff $f_j(\rho_{max}) \leq 0$. In such circumstances

$$\begin{aligned} f_j(\rho_{max}) = \bar{S}^h w\bar{D}_1(1-p)^2 & \frac{\bar{D}_1(1-p)^2 + 8h - \sqrt{\bar{D}_1^2(1-p)^4 + 16\bar{D}_1 h(1-p)^2}}{16h^2} \\ & - (\bar{D}_1 + \bar{D}_j)(1-p) + \sqrt{\bar{D}_j^2 - \bar{D}_1\bar{D}_j}(1-p) \leq 0. \end{aligned} \tag{C.14}$$

Since $f_1(\rho_{max}) \leq f_j(\rho_{max})$, with equal condition $\bar{D}_1 = \bar{D}_j$, we can find that the interior equilibrium in pure HV situation can be reached when either one of the prerequisites is satisfied.

(i) $\frac{16h^2[(\bar{D}_1+\bar{D}_j)(1-p)+\sqrt{\bar{D}_j^2-\bar{D}_1\bar{D}_j}(1-p)]}{D_1(1-p)^2[\bar{D}_1(1-p)^2+8h-\sqrt{\bar{D}_1^2(1-p)^4+16\bar{D}_1 h(1-p)^2}]} < \bar{S}^h w.$

(ii) $\frac{2h}{1-p} < \bar{S}^h w < \frac{16h^2[(\bar{D}_1+\bar{D}_j)(1-p)-\sqrt{\bar{D}_j^2-\bar{D}_1\bar{D}_j}(1-p)]}{D_1(1-p)^2[\bar{D}_1(1-p)^2+8h-\sqrt{\bar{D}_1^2(1-p)^4+16\bar{D}_1 h(1-p)^2}]}.$

j is the index of second smallest $\bar{D}_.$.

In the second part, we find out the optimal decision for the platform. Because w and p have implicit forms, it is intuitive for us to transform other parameters into expressions merely containing them. Eq. (12) now is given by:

$$\max_{p,w} \Pi_p = \sum_i D_i p - \sum_i R_i S_i^h. \tag{C.15}$$

Based on the method of Lagrangian Multiplier, we set

$$L = \Pi_p - \lambda(\rho - \frac{\sum_i S_i}{\bar{S}^h w}), \tag{C.16}$$

and calculate $\frac{\partial L}{\partial p} = 0$, $\frac{\partial L}{\partial w} = 0$, $\frac{\partial L}{\partial \rho} = 0$ and $\frac{\partial L}{\partial \lambda} = 0$:

$$\frac{\partial L}{\partial p} = \rho[\sum_i S_i + (p-w)(\sum_i S'_{ip})] + \lambda(\frac{\sum_i S'_{ip}}{\bar{S}^h w}) = 0, \tag{C.17}$$

$$\frac{\partial L}{\partial w} = -\rho(\sum_i S_i) - \lambda\frac{\sum_i S_i}{\bar{S}^h w^2} = 0, \tag{C.18}$$

$$\frac{\partial L}{\partial \rho} = (\sum_i S_i)(p-w) + \rho(p-w)(\sum_i S'_{i\rho}) - \lambda(1 - \frac{\sum_i S'_{i\rho}}{\bar{S}^h w}) = 0, \tag{C.19}$$

$$\frac{\partial L}{\partial \lambda} = \rho - \frac{\sum_i S_i}{\bar{S}^h w} = 0, \tag{C.20}$$

where $S'_{\{i\rho,ip\}}$ represents the partial derivative of S_i to ρ and p.

Plugging Eqs. (C.18) and (C.20) into Eqs. (C.17) and (C.19), a pair of expressions symmetric about i can be found:

$$\sum_i pS'_{ip} = \sum_i \rho S'_{i\rho}, \tag{C.21}$$

$$\sum_i S_i + S'_{ip}(p-2w) = 0. \tag{C.22}$$

Notice that both (C.21) and (C.22) are reversible and consecutive about p and ρ, so if they do not have a common solution on either of the regions i, we can claim no common solution on the sum of all regions as a convex hull.

Through some transformation of Eqs. (C.21) and (C.22),

$$\bar{D}_i(2p-1)^2 - \bar{D}_i(1-p)(2p-1)\frac{\rho}{1-\rho} + \frac{h\rho^4}{(1-\rho)^3} = 0, \tag{C.23}$$

$$\frac{4h}{1-\rho}(\rho^2 + \frac{\bar{D}_i}{\bar{S}^h})^2 - \bar{D}_i\rho^2(1-2p) - \frac{2\bar{D}_i^2(1-p)}{\bar{S}^h} = 0, \tag{C.24}$$

has 2 common solutions resonating with $\pm$ regarded as $+$ or $-$ with $\rho > 0$ if $\bar{S} > 4h$. A straightforward judgment is from the scale of $\bar{S}$, as $\lim_{\bar{S}\to 4h} D_i = 0, \lim_{\bar{S}\to\infty} D_i = \bar{D}_i(1-p)$. Also Π_p is smooth and $\Pi_p|_{\rho=0} = 0$, $\Pi_p|_{\rho=\rho*} = \rho^*(\sum_i S_i)(p^* - \frac{h\rho^*}{(1-\rho^{*2})\bar{S}}) \geq 0$, hence the unique extremum turns out to be maximum. Consequently, the divide of $\bar{S}$, noted as $\bar{S}_0$, should guarantee p^* and ρ^* to satisfy Eqs. (C.23), (C.24) and

$$\bar{D}^2(1-p)^2 - \frac{4\bar{D}h\rho^2}{1-\rho} = 0. \tag{C.25}$$

Numerically solving Eqs. (C.23) and (C.24) is impossible. We adopt an approximation eliminating the ρ^4 within them on ρ^*:

$$\rho_+^* \approx \frac{\sqrt{(3\bar{D}_i - \frac{4h\bar{D}_i}{\bar{S}})^2 - (80h-4)(\frac{8h\bar{D}_i}{\bar{S}} - 2\bar{D}_i)} - 3\bar{D}_i + \frac{4h\bar{D}_i}{\bar{S}}}{40h-2}, \tag{C.26}$$

$$\rho_-^* \approx \frac{\sqrt{(\bar{D}_i - \frac{\bar{D}_i h}{\bar{S}})^2 - (6h-\bar{D}_i)(\frac{4h\bar{D}_i}{\bar{S}} - \bar{D}_i)} - \bar{D}_i + \frac{h\bar{D}_i}{\bar{S}}}{6h-\bar{D}_i}. \tag{C.27}$$

Notice that $\rho_+^* < \rho_-^*$, and the actual ρ^* locates in between. These approximations are highly precise when $\bar{S}$ is small. When $\bar{S}$ is extremely large so that $\rho^4 \gg \frac{\bar{D}_i}{\bar{S}}$, real ρ^* might be smaller than ρ_+^*. Under such a circumstance, $\frac{1}{2} + O(\rho^2) < p < \frac{1}{2}O(\rho)$, and thus $\rho_+^* = O(\frac{1}{\bar{S}^4})$. Correspondingly,

$$p_\pm^* = \frac{1}{2} + \frac{\bar{D}^2 - \frac{4h\bar{S}}{1-\rho_\pm^*}({\rho_\pm^*}^2 + \frac{\bar{D}}{\bar{S}})^2}{2\bar{S}\bar{D}\rho^* - \pm^2 + 2\bar{D}^2}, \tag{C.28}$$

$$w_\pm^* = \frac{\pm\sqrt{\bar{D}^2(1-p_\pm^*)^2 - \frac{4\bar{D}h{\rho_\pm^*}^2}{1-\rho_\pm^*}} + \bar{D}(1-p_\pm^*)}{2{\rho_\pm^*}^2\bar{S}}, \tag{C.29}$$

p^* has a loose but explicit boundary: $\frac{1}{2} < p^* < \frac{1}{2} + \frac{\sqrt{(\bar{D}_i - \frac{\bar{D}_i h}{\bar{S}})^2 - (6h-\bar{D}_i)(\frac{4h\bar{D}_i}{\bar{S}} - \bar{D}_i)} - \bar{D}_i + \frac{h\bar{D}_i}{\bar{S}}}{24h - 5\bar{D}_i - 2\sqrt{(\bar{D}_i - \frac{\bar{D}_i h}{\bar{S}})^2 - (6h-\bar{D}_i)(\frac{4h\bar{D}_i}{\bar{S}} - \bar{D}_i)} + \frac{h\bar{D}_i}{\bar{S}}}$. From

$$\frac{\partial p^*}{\partial \bar{D}_i} = \frac{E'_\rho F'_{\bar{D}_i} - E'_{\bar{D}_i} F'_\rho}{E'_p F'_\rho - E'_\rho F'_p} > 0, \tag{C.30}$$

p^* increases in $\bar{D}_i$. Similarly, p^* decreases in h. However, $\frac{\partial^2 p^*}{\partial \bar{S}^2} < 0$, so p^* increases and then decreases in $\bar{S}$. Analogous results can be formed in terms of ρ^*, as from Eq. (C.23), $\frac{\partial p^*}{\partial \rho^*} < 0$. Notice that $\frac{\partial p^*}{\partial \bar{S}}|_{\bar{S}=\bar{S}_0}$ cannot be easily determined, so the turning point of monotonicity is irrelevant with $\bar{S}_0$. As for the HV drivers' total wage, when $\bar{S} > \bar{S}_0$, w decreases with p and ρ, and when $\bar{S} < \bar{S}_0$, w increases with p and ρ. Recall the monotonicity of p, ρ, and the turning point, we can get that w^* increases in $\bar{D}_i$, h, and decreases in $\bar{S}$. It is also accessible from $\frac{\partial \Pi_p}{\partial \bar{S}}$ that Π_p^* increases in $\bar{S}$ and decreases in h. □

Proof of Corollary 1. We notice that when we fix $\bar{D}_j$ to every $\bar{D}_{j0}$, Π_p reaches a maximum at $\bar{D}_i$ when $\bar{D}_i = \bar{D}_{j0}$. Also if $\bar{D}_i = \bar{D}_j$, through $\frac{\partial \Pi_p}{\partial \bar{D}_i}$ we can see that Π_p increases in $\bar{D}_i$. □

Proof of Proposition 2. Without loss of generality, we use the case of two regions to finish the rest proof. The proof of the general case can follow the same procedure in proving Proposition 1.

We apply $S_1^a = \gamma S^a$ and $S_2^a = (1-\gamma)S^a$ into Eq. (16) to get:

$$\Pi_p(p, S^a, \gamma) = \frac{1}{2}(B_1 + B_2 - \sqrt{B_1^2 - 4C_1} - \sqrt{B_2^2 - 4C_2})(p - c_v) - S^a c_f, \tag{C.31}$$

where B_i, C_i are parameters related to(p, S^a, γ). Calculate up to the second derivative we can get:

$$\begin{aligned}\frac{\partial \Pi_p}{\partial p} =& \frac{1}{2}(S^a + (1-2p+c_v)(\bar{D}_1 + \bar{D}_2) + \frac{\bar{D}_1 h}{\gamma S^a} + \frac{\bar{D}_2 h}{(1-\gamma)S^a} \\ &+ \sum_{i=1}^{2} \frac{(p-c_v)(\bar{D}_i^2(1-p) + \frac{\bar{D}_i^2 h}{S_i} - S_i\bar{D}_i) - B_i^2 + 4C_i}{\sqrt{B_i^2 - 4C_i}}),\end{aligned} \tag{C.32}$$

$$\begin{aligned}\frac{\partial \Pi_p}{\partial S^a} =& \frac{1}{2}(p-c_v)(1 - \frac{\bar{D}_1 h}{\gamma S_a^2} - \frac{\bar{D}_2 h}{(1-\gamma)S_a^2} - \frac{B_1(\gamma - \frac{\bar{D}_1 h}{S_a^2\gamma}) - 2\bar{D}_1(1-p)\gamma}{\sqrt{B_1^2 - 4C_1}} \\ &- \frac{B_2((1-\gamma) - \frac{\bar{D}_2 h}{S_a^2(1-\gamma)}) - 2\bar{D}_2(1-p)(1-\gamma)}{\sqrt{B_2^2 - 4C_2}}) - c_f,\end{aligned} \tag{C.33}$$

$$\frac{\partial \Pi_p}{\partial \gamma} = \frac{1}{2}(p-c_v)((-\frac{\bar{D}_1 h}{S^a \gamma^2} + \frac{\bar{D}_2 h}{S^a(1-\gamma)^2}) - \frac{B_1(S^a - \frac{\bar{D}_1 h}{S^a \gamma^2}) - 2\bar{D}_1(1-p)S^a}{\sqrt{B_1^2 - 4C_1}} - \frac{B_2(-S^a - \frac{\bar{D}_2 h}{S^a(1-\gamma)^2}) + 2\bar{D}_2(1-p)S^a}{\sqrt{B_2^2 - 4C_2}}). \tag{C.34}$$

From Eq. (C.32) we can conclude that if $c_v \geq 1$, then $\frac{\partial \Pi_p}{\partial p} > 0$, and thus $p^* = 1$. Otherwise, firstly to maintain it possible for all derivatives to reach zero p^* must be greater than $\max\{\frac{1}{2}, c_v + c_f\}$. Secondly because $\frac{\partial^2 \Pi_p}{\partial p^2} < 0$ there still exists a unique $p^* \in (\max\{\frac{1}{2}, c_v + c_f\}, 1)$ such that $\frac{\partial \Pi}{\partial p}(p^*) = 0$, maximizing Π_p. Thirdly by

$$\frac{\partial p^*}{\partial c_v} = -\frac{\partial^2 \Pi_p^2}{\partial p^2} / \frac{\partial \Pi_p^2}{\partial p \partial c_v}, \tag{C.35}$$

p^* increases in c_v. However, the monotonicity of p in terms of $\bar{D}_i$ and h is strongly influenced by the actual value of S_a^* and γ^*, so it is too complex to determine with only proof of the existence.

Also notice that

$$\frac{\partial \Pi_p}{\partial S_a}|_{S^a=0} = \frac{\partial \Pi_p}{\partial S_a}|_{S^a=\infty} = -c_f, \tag{C.36}$$

but

$$\frac{\partial \Pi_p}{\partial S_a}|_{S^a=\sqrt{\bar{D}_i h}} = \frac{1}{2}(p-c_v)(1 - \frac{1}{\gamma} - \frac{1}{(1-\gamma)} - \frac{(\gamma - \frac{1}{\gamma})(2\sqrt{\bar{D}_1 h} + \bar{D}_1(1-p)) - 2\gamma \bar{D}_1(1-p)}{|2\sqrt{\bar{D}_1 h} - \bar{D}_1(1-p)|} - \frac{((1-\gamma) - \frac{1}{1-\gamma})(2\sqrt{\bar{D}_2 h} + \bar{D}_2(1-p)) - 2(1-\gamma)\bar{D}_2(1-p)}{|2\sqrt{\bar{D}_2 h} - \bar{D}_2(1-p)|}) \tag{C.37}$$

can be positive, if $\bar{D}_i$, h and c_v are set subtly (for instance, let $\bar{D}_1 = \bar{D}_2 = 1000$, $h = 200$, $c_v = 0.16$, $c_f = 0.05$). Therefore, some particular

$$c_{f0}(\bar{D}_i, h, c_v) = \max_{S^a} \frac{\partial \Pi_p}{\partial S^a} + c_f \tag{C.38}$$

exists, which increases in $\bar{D}_i$ and decreases in h and c_v, exists and while $c_f \geq c_{f0}$, $\frac{\partial \Pi}{\partial S_a} < 0$ and $S_a^* = 0$. Otherwise $S^a > 0$ exists. Fixing p and γ, as

$$1 - \frac{\bar{D}_1 h}{\gamma S_a^2} - \frac{\bar{D}_2 h}{(1-\gamma)S_a^2} - \frac{B_1(\gamma - \frac{\bar{D}_1 h}{S_a^2 \gamma}) - 2\bar{D}_1(1-p)\gamma}{\sqrt{B_1^2 - 4C_1}} - \frac{B_2((1-\gamma) - \frac{\bar{D}_2 h}{S_a^2(1-\gamma)}) - 2\bar{D}_2(1-p)(1-\gamma)}{\sqrt{B_2^2 - 4C_2}} \tag{C.39}$$

strictly decreases with S_a, it can be observed that S_a^* increases in $\bar{D}_i$, h and decreases in c_v and c_f.

From the symmetrical characteristic of Eq. (C.34) we can see that when $\bar{D}_1 = \bar{D}_2$, no matter how other exogenous parameters are varying, $\gamma = \frac{1}{2}$ is a solution of $\frac{\partial \Pi_p}{\partial \gamma} = 0$, and this extremum increases with $\bar{D}_1$ and decreases with $\bar{D}_2$. However, Eq. (C.34) has at most 3 different solutions depending on the value of p^*, S_a^*, $\bar{D}_i$, and h, so we need to analyze all situations individually.

As $p^* > c_v$, we are able to find $\frac{\partial^2 \Pi_p}{\partial \gamma^2}$ quadratic, so when $\bar{D}_1 = \bar{D}_2$, four different situations can be depicted:

1. when $h = 0$, resulting $\frac{\partial^2 \Pi_p}{\partial \gamma^2} < 0$ on $(0, 1)$, $\gamma = \frac{1}{2}$ is the maximum point of Π_p;
2. when h is small compared to $\bar{D}_i$ so that there exist 3 solutions for $\frac{\partial \Pi_p}{\partial \gamma} = 0$, but $\Pi_p|_{\gamma=0} = \Pi_p|_{\gamma=1} < \Pi_p|_{\gamma=\frac{1}{2}}$, $\gamma = \frac{1}{2}$ is also the maximum point;
3. when h is relatively large so that $\Pi_p|_{\gamma=0} = \Pi_p|_{\gamma=1}$ are larger than $\Pi_p|_{\gamma=\frac{1}{2}}$, while $\frac{\partial^2 \Pi_p}{\partial \gamma^2}$ still have non-positive intervals, $\gamma^* = 0$ and $\gamma^* = 1$ are maximum points;
4. when h is large enough so that $\frac{\partial^2 \Pi_p}{\partial \gamma^2} > 0$, $\gamma^* = 0$ and $\gamma^* = 1$ are two maximum points.

So we can conclude that when $\bar{D}_1 = \bar{D}_2$, there exists some h_0 related to $\bar{D}_i$ and if $h < h_0$, $\gamma^* = \frac{1}{2}$, else $\gamma^* = 0$ or 1.

When it comes to the circumstance where $\bar{D}_1 \neq \bar{D}_2$ (we set $\bar{D}_1 < \bar{D}_2$ without losing generality), $\Pi_p|_{\gamma=0} > \Pi_p|_{\gamma=1}$ and Eq. (C.34) has more varieties depending on the relationship with $\bar{D}_i$ and h. From the previous conclusions, two distinct categories of h can be set: one where h is so large that $\frac{\partial^2 \Pi_p}{\partial \gamma^2} > 0$ and the other that $\frac{\partial^2 \Pi_p}{\partial \gamma^2}$ has non-positive intervals. In the first category, the solution of $\frac{\partial \Pi_p}{\partial \gamma} = 0$ is just a possible minimum point competing with $\gamma = 1$, and $\gamma = 0$ is the maximum point. In the second one, it gets more complex:

1. while $\bar{D}_1$ is much smaller than $\bar{D}_2$, according to the features of $\frac{\partial^2 \Pi_p}{\partial \gamma^2}$, Eq. (C.34) has only one zero point, which shares the same feature as the first category;

2. while $\bar{D}_1$ is larger so that Eq. (C.34) has 3 solutions, but the accumulation of the positive part does not neutralize the negative, γ^* is still 0;
3. while $\bar{D}_i$ is large enough, making the extremum exceed $\Pi_p|_{\gamma=0}$, the intermediate solution turns out to be maximum, so a proportion $t_0(h)$ exists and while $t = \frac{\bar{D}_1}{\bar{D}_2} > t_0$, γ^* strictly locates in $(0, 1)$.

Though γ^* does not change with c_v when $p*$ and S_a^* are fixed, γ^* changes in the same direction when p or S_a increases while they change in the opposite directions when c_v is increasing. So its monotonicity in terms of c_v is too complex to analyze, and so is h. Adding the symmetric case where $\bar{D}_1 > \bar{D}_2$, and combining with the limitations of $p*$ and S_a^*, the final proposition is formed.

Summary results.

1. As long as $c_v < 1$, $\max\{\frac{1}{2}, c_v + c_f\} < p^* < 1$ exists and increases in c_v.
2. There exists a $\hat{c}_f(\bar{D}_i, h, c_v)$, and as long as $\hat{c}_f < \hat{c}_f$, $S_a^* > 0$ exists and decreases in c_f. $\hat{c}_f$ increases in $\bar{D}_i$ and decreases in h and c_v, S_a^* increases in $\bar{D}_i$, h and decreases in c_v and c_f.
3. There exists an $h_0(\bar{D}_1, \bar{D}_2)$ and $t_0(h) < 1$ and only if $h < h_0$ and $t_0 < t = \frac{\bar{D}_1}{\bar{D}_2} < \frac{1}{t_0}$ are satisfied can γ^* locate in $(0, 1)$. In such cases, γ^* increases in $\bar{D}_1$ and decreases in $\bar{D}_2$.

Under these conditions, it can be claimed that a unique combination of (p, S^a, γ) exists, maximizing Π_p. The changing directions of S_a^* and γ^* with other parameters are too trivial to be highlighted in Proposition 2. □

Proof of Proposition 3. Without loss of generality, we use the case of two regions to finish the rest proof. The proof of the general case can follow the same procedure in proving Proposition 1.

In this part we mark:

$$\rho = \frac{D_1}{S_1^h + S_1^a} = \frac{D_2}{S_2^h + S_2^a}, \tag{C.40}$$

$$Q^{\pm} = \pm\sqrt{\bar{D}_1^2(1-p)^2 - 4\frac{\bar{D}_1 h\rho^2}{1-\rho}} \pm \sqrt{\bar{D}_2^2(1-p)^2 - 4\frac{\bar{D}_2 h\rho^2}{1-\rho}} + (\bar{D}_1 + \bar{D}_2)(1-p). \tag{C.41}$$

From interior Wardrop equilibrium (Eq. (8) to Eq. (11)), the exact form of the number and wage of HV drivers can be acquired:

$$S_i^h = \frac{\pm\sqrt{\bar{D}_i^2(1-p)^2 - 4\frac{\bar{D}_i h\rho^2}{1-\rho}} + \bar{D}_i(1-p)}{2\rho} - S_i^a, \tag{C.42}$$

$$w = \frac{S_1^h + S_2^h}{\bar{S}^h\rho} = \frac{Q}{2\rho^2\bar{S}^h} - \frac{S^a}{\rho\bar{S}^h}. \tag{C.43}$$

The original optimization problem with p, w, S^a, γ can be transformed into another one with p, ρ, S^a, γ:

$$\max_{p,\rho,S^a,\gamma} \Pi_p = -\frac{Q^2}{4\rho^2\bar{S}^h} + \frac{Qp}{2} + \frac{QS^a}{\bar{S}^h\rho} - \frac{S_a^2}{\bar{S}^h} - S^a\rho c_v - S^a c_f, \tag{C.44}$$

γ does not appear in the final expression, remaining 3 variables. The monotonicity of S^a is quite simple as $\frac{\partial^2\Pi_p}{\partial S_a^2} < 0$. So if

$$\frac{Q^*}{\bar{S}^h\rho^*} - \rho^* c_v - c_f \le 0, \tag{C.45}$$

then Π_p decreases with S^a, $S_a^* = 0$, and the problem deteriorates to pure HV market. Otherwise

$$\Pi_p^{S_a^*} = \Pi_p|_{S_a = \frac{Q^*}{2\rho^*} - \frac{\bar{S}^h\rho^* c_v + \bar{S}^h c_f}{2}} = \frac{\bar{S}^h}{4}\left(\frac{Q^*}{\bar{S}^h\rho^*} - \rho^* c_v - c_f\right)^2 + \frac{Q^* p^*}{2} - \frac{Q^{*2}}{4\rho^{*2}\bar{S}^h}. \tag{C.46}$$

An obvious result is that when $\bar{S}^h$, c_v, and c_f are large, there will be no necessity for the platform to distribute AV transportation. The wages of HV drivers are $\frac{Q}{2\rho^2\bar{S}^h}$ and $\frac{(\rho c_v + c_f)}{2\rho}$ before and after the introduction of AV, which indicates that as the total amount of HVs shrinks or the cost of AVs grows, the wage of HV drivers witnesses an increase until the platform decides to put AVs into use. That is, compared to a pure HV market, the platform reduces the wage by introducing an AV fleet.

From Eq. (C.46), the profit of a mixed market with both AV and HV is always equal to or larger than a pure HV market. Moreover, the optimizer p, ρ shall satisfy:

$$\frac{\partial\Pi_p^{S_a^*\neq 0}}{\partial p} = \frac{Q'_p p}{2} + \frac{Q}{2} - \frac{Q'_p}{2\rho}(\rho c_v + c_f) = 0, \tag{C.47}$$

$$\frac{\partial\Pi_p^{S_a^*\neq 0}}{\partial \rho} = \left(-\frac{Q'_\rho\rho - Q}{2\rho^2} + \frac{\bar{S}^h c_v}{2}\right)(\rho c_v + c_f) - \frac{Qc_v}{2\rho} + \frac{Q'_\rho p}{2} = 0, \tag{C.48}$$

while

$$\frac{\partial \Pi_p^{S_a^*=0}}{\partial p} = \frac{Q'_p p}{2} + \frac{Q}{2} - \frac{QQ'_p}{2\rho^2 \bar{S}^h} = 0, \tag{C.49}$$

$$\frac{\partial \Pi_p^{S_a^*=0}}{\partial \rho} = -\frac{QQ'_\rho \rho - Q^2}{2\rho^3 \bar{S}^h} + \frac{Q'_\rho p}{2} = 0. \tag{C.50}$$

From the derivatives, if $\frac{Q'_p}{2\rho}(\frac{Q}{\bar{S}^h \rho} - \rho c_v - c_f) > 0$, then the optimal p^* is larger for a mixed market than a pure market. Similarly, if $\frac{Q'_\rho \rho - Q - \rho^2 \bar{S}^h c_v}{2\rho^2}(\frac{Q}{\bar{S}^h \rho} - \rho c_v - c_f) > 0$, then the optimal ρ^* is larger for a mixed market than a pure market. Q^+ decreases with both p and ρ. Thus if the maximum point of Π_p is acquired on Q^+, both optimal p^* and ρ^* are always larger with a pure market. In contrast, Q^- increases with both p and ρ, so the optimal p^* is smaller with a pure market, while ρ^* cannot be easily compared: If $\bar{D}_i$ are large or $\bar{S}^h$ is small, the optimal ρ^* is more likely to be smaller with a pure market.

The maximum points might not be accessible. To maintain In Eq. (C.45) for Q^+, p and ρ needs to be as large as possible. The maximum p_{max} and ρ_{max} shall satisfy $\bar{D}_i^2(1-p_{max})^2 - 4\frac{\bar{D}_i h \rho_{max}^2}{1-\rho_{max}} = 0$. Therefore, $\bar{D}_i$ must be larger than a threshold $\underline{d}_i(\bar{S}^h, c_f, c_v)$. Symmetric value $\bar{d}_i(\bar{S}^h, c_f, c_v)$ should not be exceeded to maintain In Eq. (C.45) for Q^-. In the meantime, to ensure the profit involving Q^+ superseding that with Q^-, h must be smaller than a threshold $h_0(\bar{S}^h, c_f, c_v)$. From In Eq. (C.45) we can also know that $\underline{d}_i(\bar{S}^h, c_f, c_v)$ increases with $\bar{S}^h$, c_v, and c_f, while $\bar{d}_i(\bar{S}^h, c_f, c_v)$ has a converse monotonicity. h_0 increases with $\bar{S}^h$ and decreases with c_v and c_f.

Now we can discuss the quantity of both drivers. When

$$\frac{Q^*}{2\rho^*} > \bar{S}^h(\rho^* c_v + c_f) \tag{C.51}$$

$S^h < S^a$. If h is extremely large or $\bar{D}_i$ is extremely small, $Q^- \to h$ and p^* satisfies $\frac{Q^*}{\bar{S}^h \rho^*} = (\rho^* c_v + c_f)$. However as $\frac{\bar{S}^h}{2}(\rho c_v + c_f) < \bar{S}^h(\rho c_v + c_f)$, p_{min} cannot satisfy In Eq. (C.51). So $S^h > S^a$. If h is extremely small or $\bar{D}_i$ is extremely large, $Q^+ \to 2\bar{D}(1-p)$, $\rho^* \to 1$ and $p^* = \frac{1+c_v+c_f}{2}$. In this case, we can also set $\bar{S}^h$ subtly so that $\frac{\bar{S}^h}{2}(\rho c_v + c_f) < \frac{Q^*}{2\rho^*} < \bar{S}^h(\rho c_v + c_f)$. So to ensure In Eq. (C.51), both $\bar{D}_i$ and h need to be moderate.

Summary results.

1. Introducing an AV fleet is always profitable for the platform. If the maintenance cost c_v and c_f are small or the supply of HV fleet is insufficient, or the total demand is within an interval $[\underline{d}_i, \bar{d}_i]$, it will be strictly profitable instead. Moreover, the interval shrinks with the increase of $\bar{S}^h$, c_v and c_f.
2. Compared to a pure HV market, w^* is reduced if the platform introduces an AV fleet. Moreover, if the delay sensitivity $h < h_0$, p^* and ρ^* decreases, otherwise p^* increases while ρ^* increases only when $\bar{D}_i$ are large or $\bar{S}^h$ is small. Additionally, h_0 increases with $\bar{S}_i^h$ and decreases with c_v and c_f.
3. There exist intervals (h_1, h_2) and $(\bar{D}_1^i, \bar{D}_i^2)$ so that when $h \in (h_1, h_2)$ or $\bar{D}^i \in (\bar{D}_1^i, \bar{D}_2^i)$, more AVs are in the market than HVs.
4. If $\rho^* < \frac{c_f}{c_v + 2c_f}$, the wage of HV drivers is higher than $c_v + c_f$, which means c_v is far smaller than c_f. □

Proof of Lemma 2. We finish three parts during the proof. (1) the calculation of equilibrium in a hybrid market. (2) the exclusion of all-or-nothing cases. (3) the calculation of equilibrium in the pure HV market.

1. **Equilibrium in hybrid market**. Since the human drivers make selfish decisions to yield a Wardrop equilibrium $R_1 = R_2$. In other words, the ratio of effective supply–demand is the same in the two regions (see Eq. (7)), so we can get $\rho_1 = \rho_2 = \rho = \frac{(1-p)\bar{D}}{w\bar{S}^h + S^a}$. Then the platform profit Π_p can be simplified as:

$$\Pi_p = (1-p)\bar{D}p - w^2 \rho \bar{S}^h - \rho c_v S^a - S^a c_f, \tag{C.52}$$

If $\rho > 1$, we can find that Π_p monotonously increases with S^a. This is because the platform can increase S^a to improve the profit without changing other decision variables (price and wage). So we just have to consider the situation $\rho < 1$. The first and second derivatives of Π_p on S^a are:

$$\frac{\partial \Pi_p}{\partial S^a} = \frac{\bar{D}(1-p)\bar{S}^h w(w - c_v)}{(S^a + \bar{S}^h w)^2} - c_f, \tag{C.53}$$

$$\frac{\partial^2 \Pi_p}{\partial S^{a2}} = \frac{2\bar{D}(1-p)\bar{S}^h w(c_v - w)}{(S^a + \bar{S}^h w)^3}. \tag{C.54}$$

There are two situations: (1) when $w \in (0, c_v)$, the function is convex, and the maximum is obtained when $S^a = 0$ or $+\infty$, which degenerates to the pure market. (2) when $w \in (c_v, 1)$, the function is concave and we can find that $\frac{\partial \Pi_p}{\partial S^a}|_{S^a=0} < 0$. That is, Π_p monotonously decreases with S^a, and the maximum is achieved at $\rho = 1$ or $S^a = 0$. Combining these findings, because we just focus on the hybrid market and thus we only have to calculate the situation given $\rho = 1$, which equals to $(1-p)\bar{D} = w\bar{S}^h + S^a$. This means we have to solve an optimization problem with two variables. The rest calculations regarding

the usual derivative are omitted. We can solve out the equilibrium results (shown in Table 3). The equilibrium AV fleet size is $S_p^{a*} = \frac{\bar{D}-\bar{c}\bar{D}-\bar{c}\bar{S}^h}{2}$, which can be positive only when $\bar{c} < \frac{\bar{D}}{\bar{D}+\bar{S}^h}$.

2. **All-or-nothing case**. We calculate the situation $R_2 \geq R_1$ and $S_1^h = 0$, namely, all human drivers work in region 2. One can observe that $R_2 \geq R_1$ equals to $\rho_2 \geq \rho_1$. Because only AVs serve in Region 1, the platform can always reduce the number of AVs in this region to improve profit. The decline of AVs in region 1 equals the increase of ρ_1, and finally, the profit reaches a maximum when $\rho_1 = \rho_2$. In other words, the platform still maintains the same ratio in two regions (i.e., $\rho_1 = \rho_2 = \rho$). As such, this situation is identical to the general one we have discussed.
3. **Equilibrium in pure HV market**. By setting $S^a = 0$, we can refine Eq. (C.52) as:

$$\Pi_p = (1-p)\bar{D}p - w^2\bar{S}^h. \tag{C.55}$$

The constraint $(1-p)\bar{D} = w\bar{S}^h$ is satisfied to maintain the supply–demand balance. The rest calculations regarding the usual derivative are omitted. The equilibrium results are shown in Table 3.

The proof of Proposition 4 can be finished by directly comparing the values in Table 3. □

Proof of Proposition 5. The calculations are identical to the ones in Lemma 2 after revising the optimization objectives. In a hybrid market, social welfare is defined as follows:

$$W = \frac{\bar{D}(1-p^2)}{2} - \frac{w^2\bar{S}^h}{2} - S^a c. \tag{C.56}$$

In a pure HV market, social welfare is defined as follows:

$$W = \frac{\bar{D}(1-p^2)}{2} - \frac{w^2\bar{S}^h}{2}. \tag{C.57}$$

The rest calculations regarding the usual derivative are omitted. The equilibrium results are shown in Table 3. The proof of Corollaries 2 and 3 can be finished by directly comparing the values. □

Proof of Lemma 3. Under a price cap ($\overline{p} < p_p^*$), we have to resolve the optimization problem in Eq. (12) with a new constraint $p < \overline{p}$. If this constraint affects the optimization process, the equilibrium will be bounded at the constraint ($p_r^* = \overline{p}$). In this light, we just have to optimize the variables S^a and w. Following the same process in proving Lemma 2, we can find the revised optimal decisions as:

$$w_{r,\overline{p}}^* = \frac{c}{2} = w_p^*, \tag{C.58}$$

$$S_{r,\overline{p}}^{a*} = \bar{D}(1-\overline{p}) - \frac{c\bar{S}^h}{2} > S_p^{a*}. \tag{C.59}$$

The social welfare can be calculated as $W_{r,\overline{p}} = \frac{1}{8}\left(3\bar{c}^2\bar{S}^h + 4\bar{D}(\overline{p}-1)(2\bar{c}-\overline{p}-1)\right)$. Under the constraint $\overline{p} < p_p^*$, we can compare $W_{r,\overline{p}}$ with W_p^*. Setting a price cap worsens social welfare only when $\bar{D} > \frac{\bar{S}^h}{2} \wedge \bar{c} > \frac{1}{3} \wedge 0 < \overline{p} < \frac{1}{2}(3\bar{c}-1)$. Otherwise, this regulation can improve social welfare. □

Proof of Lemma 4. Following the same process in proving Lemma 3, we can find the revised optimal decisions as:

$$p_{r,\underline{w}}^* = \frac{1+c}{2} = p_p^*, \tag{C.60}$$

$$S_{r,\underline{w}}^{a*} = \frac{1}{2}(-\bar{c}\bar{D} + \bar{D} - 2\bar{S}^h\underline{w}) < S_p^{a*}. \tag{C.61}$$

The social welfare can be calculated as $W_{r,\underline{w}} = \frac{3}{8}(\bar{c}-1)^2\bar{D} + c\bar{S}\underline{w} - \frac{\bar{S}^h\underline{w}^2}{2}$. Under the constraint $\underline{w} > w_p^*$, we can compare $W_{r,\underline{w}}$ with W_p^*. Setting a minimum wage worsens social welfare only when $\bar{c} < \min\{\frac{2}{3}, \frac{\bar{D}}{\bar{D}+\bar{S}^h}\} \wedge \underline{w} > \frac{3\bar{c}}{2}$. Otherwise, this regulation can improve social welfare. □

Proof of Lemma 5. One can observe that the control of maximum fleet size $\overline{S}$ equals the restriction of minimum price standard when the platform reaches a supply–demand balance. We can directly investigate if there exists a minimum price standard $\underline{p} > p_p^*$.

$$w_{r,\underline{p}}^* = \frac{c}{2} = w_p^*, \tag{C.62}$$

$$S_{r,\underline{p}}^{a*} = \bar{D}(1-\underline{p}) - \frac{c\bar{S}^h}{2} < S_p^{a*}. \tag{C.63}$$

The social welfare can be calculated as $W_{r,\underline{p}} = \frac{1}{8}\left(3\bar{c}^2\bar{S}^h + 4\bar{D}(\underline{p}-1)(2\bar{c}-\underline{p}-1)\right)$. Under the constraint $\underline{p} > p_p^*$, we can observe that $W_{r,\underline{p}}$ can never dominate W_p^* such that the control of fleet size worsens social welfare. The finding holds in our stylized model because we only consider the ride-sourcing market. Nevertheless, this regulation works effectively in reality since it can relieve the traffic jam in the network in most cases. □

Proof of Lemma 6. Following the same process in proving Lemma 3, we can find the revised optimal decisions under minimum size control as:

$$p^*_{r,\underline{S}^a} = \frac{2\bar{D} - 2\underline{S}^a + \bar{S}^h}{2(\bar{D} + \bar{S}^h)} < p^*_p, \tag{C.64}$$

$$w^*_{r,\underline{S}^a} = \frac{\bar{D} - 2\underline{S}^a}{2(d + \bar{S}^h)} < w^*_p. \tag{C.65}$$

The social welfare can be calculated as $W_{r,\underline{S}^a} = \frac{8(1-\bar{c})\bar{D}\underline{S}^a - 4\underline{S}^a(2\bar{c}\bar{S}^h + \underline{S}^a) + 3\bar{D}\bar{S}^h}{8(\bar{D}+\bar{S}^h)}$. Under the constraint $S^{a*}_p < \underline{S}^a$, we can compare $W_{r,\underline{S}^a}$ with W^*_p. Controlling the minimum size of the AV fleet worsens social welfare only when $\bar{c} > \frac{2\bar{D}}{3\bar{D}+3\bar{S}^h} \wedge \underline{S}^a > \frac{3}{2}(\bar{D} - \bar{c}\bar{D} - \bar{c}\bar{S}^h)$. Otherwise, this regulation can improve social welfare. Similarly, we can examine whether the maximum control of the AV fleet (i.e., $S^a < \overline{S}^a$) can benefit social welfare. The result indicates that this regulation can never improve social welfare. □

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